Constant oracle-risk bounds for normalized spectral likelihood

End-to-end Bernoulli SBM and degree-corrected SBM results

Research proof note for the spectral-lrt project

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Abstract

We give finite, implementable classifiers based only on retained normalized spectral information and prove expected misclassification risk at most a constant times a precisely specified local oracle risk. For ordinary SBM, the signed Euclidean spectral reconstruction and finite growing/replacement likelihood initializer are covered on an explicit nontrivial model class with $np \geq 5 \log n$, with a broader information–degree margin extension. For bounded-degree-parameter DC-SBM, a degree-metric spectral reconstruction, deterministic spectral initialization, estimated nuisance parameters and Bernoulli profile likelihood give the bound for general positive full-rank block profiles when $C_0 \log n \leq np$ and $np^2 \leq C_1$. The reference is the full-edge profile-LRT oracle with true neighbor parameters. Equal block exposures further imply a bound relative to the oracle supplied with the target's true degree parameter. A sharp comparison explicitly identifies this reference with the classical balanced planted-partition DC-SBM local oracle; that oracle comparison requires only $np \to \infty$ and bounded np^2. A two-point counterexample outside the classical assortative class shows that the stronger reference cannot be matched throughout arbitrary DC-SBM. The proof keeps the sharp $d^{-1/2}$ prefactor, establishes actual full-graph-to-LOO score control, and pays every failure event at the oracle scale. The algorithm modifications and model restrictions are explicit; a bounded risk ratio is not a claim that the ratio converges. The final section proves degree-floor geometry and an exponential-moment transfer lemma below logarithmic degree, and states the remaining algorithmic estimates without claiming a completed sublogarithmic theorem.

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1 Models, oracles, algorithms, and theorems

The objective throughout is the literal inequality

$$\mathbb{E}\ell(\widehat{z}, z) \le CR_*(n), \qquad C < \infty \text{ independent of } n. \tag{1}$$

An estimate of the form $\exp\{-(1 - o(1))I_n\}$ does not establish (1) when the oracle error is of order $d^{-1/2}\exp(-I_n)$. The proof below controls bounded changes in a tilted cumulant and retains that prefactor. For comparison, the main theorem of Zhang [5] characterizes the leading error exponent of a spectral clustering algorithm, including a polynomial exceptional-event term. We use its distinction between an actual spectral algorithm and its population surrogate, while proving a different likelihood comparison on the parameter ranges stated here.

1.1 Bernoulli models and fixed structural constants

There are n vertices and $K \ge 2$ groups, with fixed K. Let $z \in [K]^n$, let Z be its indicator matrix, and let n_a be the size of group a. Throughout $n_a \ge \pi_- n$ for a fixed $\pi_- > 0$. The edges above the diagonal are independent, the adjacency matrix A is symmetric, and $A_{ii} = 0$. In DC-SBM,

$$A_{ij} \sim \mathrm{Bern}(\theta_i\theta_j pC_{z_i z_j}), \quad i < j, \qquad d = np, \qquad \Omega = \Theta Z(pC)Z^\top\Theta. \tag{2}$$

Here $\Theta = \mathrm{diag}(\theta_i)$, $C = C^\top$, and the fixed structural bounds are

$$0 < \theta_- \le \theta_i \le \theta_+ < \infty, \quad \sum_{i:z_i=a} \theta_i = n_a, \quad 0 < c_- \le C_{ab} \le c_+ < \infty, \quad \sigma_{\min}(C) \ge \kappa > 0. \tag{3}$$

The model class is nonempty only when $1 \in [\theta_-, \theta_+]$. All constants in asymptotic bounds may depend on these structural bounds, but not on n. One may take a fixed C, or a sequence in this same compact class. Ordinary SBM is the special case $\theta_i = 1$, with block probability matrix $P = pC$.

The completed matrix Ω includes a diagonal for algebraic convenience; the actual mean is $\mathbb{E}A = \Omega - \mathrm{diag}(\Omega_{ii})$. That deterministic difference has norm $O(p)$ and is included in all error bounds. Put $D = \mathrm{diag}(A\mathbf{1})$ and $\lambda_i = (\mathbb{E}A\mathbf{1})_i$. Thus $\lambda_i \asymp d$ and $\Omega\mathbf{1}$ equals $d\theta_i S_{z_i}$ in coordinate i, where

$$\pi_a = n_a/n, \qquad S_a = \sum_b \pi_b C_{ab}. \tag{4}$$

The loss for the estimated partition is

$$\ell(\widehat{z}, z) = \frac{1}{n}\min_{\sigma\in\mathfrak{S}_K}\sum_i \mathbf{1}\{\sigma(\widehat{z}_i) \ne z_i\}.$$

1.2 Three explicitly different local oracles

All oracle classifiers below use the raw incident row $A_{i,-i}$ and the true neighbor labels. They are local likelihood classifiers: an exact global block-size constraint is not used to determine the one missing label. For simplicity candidate weights are uniform. Any fixed weights bounded above and below by positive constants give the same conclusions.

For ordinary SBM, the oracle chooses the maximizer of

$$\ell_{i,*}(a) = \sum_{j\ne i}\{A_{ij}\log P_{az_j} + (1 - A_{ij})\log(1 - P_{az_j})\}. \tag{5}$$

Its average error probability is $R_{\mathrm{SBM},*}$. For DC-SBM, the *known-target-degree oracle* replaces P_{az_j} in (5) by $\theta_i\theta_j pC_{az_j}$ and is supplied with every true θ_j, including θ_i. Its risk is $R_{\theta,*}$.

The *profile oracle* is supplied with true neighbor parameters and pC, but profiles the target degree separately for every candidate:

$$\ell_{i,\mathrm{prof},*}(a) = \sup_{t\geq 0} \sum_{j\neq i} \{A_{ij} \log(t\theta_j pC_{az_j}) + (1 - A_{ij}) \log(1 - t\theta_j pC_{az_j})\}. \tag{6}$$

We restrict candidate edge probabilities to at most $q_{\max} = 1/2$; the constraint is inactive on the events used in the sparse theorem. Write $R_{\mathrm{prof},*}$ for its average error probability. This is a full-edge generalized likelihood-ratio reference. It is not described as a Bayes test between two fixed known-θ_i laws. In all three definitions, the risk averages the actual directional error at each fixed true vertex. Ties are resolved by a fixed ordering; strict-tail arguments below cover any such ordering.

1.3 One encoder and two decoders

When all degrees are positive, form

$$S = D^{-1/2} A D^{-1/2} = I - L.$$

Retain the degree eigenvector $D^{1/2}\mathbf{1}/\left\|D^{1/2}\mathbf{1}\right\|_2$ and the $K-1$ other eigenvectors with largest absolute eigenvalues in its orthogonal complement. Let $U^\top U = I_K$ and $SU = U\Lambda$, where $\Lambda = \mathrm{diag}(\mu_1, \ldots, \mu_K)$. On the proved event this is exactly the informative K-dimensional space. The encoder supplies only (D, U, Λ) to the decoder. Put

$$X = D^{-1/2}U, \qquad AX = D^{1/2}U\Lambda. \tag{7}$$

Thus row actions on the retained space require no later edge lookup. The normalized Laplacian eigenvalues contain the same information via $\mu = 1 - \lambda_L$. Exact arithmetic and algebraic retained spaces define the theorem; a fixed numerical rank threshold is not its definition.

The ordinary-SBM decoder uses the *signed* reconstruction

$$Q = \mathrm{orth}(X), \quad \Pi_X = QQ^\top, \quad R_E = A\Pi_X. \tag{8}$$

It fits free Bernoulli row profiles with the finite growing/replacement schedule and mandatory final E/M step specified in section 4. Only probabilities are clipped; R_E is not clipped. Neither true labels nor a correct initial partition are inputs.

For DC-SBM we make the explicit degree-metric correction

$$H_D = D^{-1/2}UU^\top D^{1/2}, \qquad R_D = AH_D = D^{1/2}U\Lambda U^\top D^{1/2}. \tag{9}$$

With $h_i = \|U_{i\cdot}\|_2^2$ and $s_i = (R_D)_{ii}/(1 - h_i)$, its hollow version is

$$R^\circ = R_D + \mathrm{diag}(s)(H_D - I). \tag{10}$$

It satisfies $\mathrm{diag}(R^\circ) = 0$, $R^\circ\mathbf{1} = D\mathbf{1}$ and $R^\circ X = AX$. It need not be symmetric after hollowing; a symmetric block parameter is fitted with the ordered-row working criterion divided by two.

For the initial partition, normalize every row of U to unit length. Repeatedly select a remaining row having the largest number of remaining rows within distance $1/8$, and remove rows within distance $1/4$ of the selected row. After K selections assign every row to the closest selected row. The separation argument in section 5 proves its $O(1/d)$ error; this is a deterministic procedure with $O(Kn^2)$ distance comparisons, not an assumed global k-means solution. Normalize degree estimates to mean one within each fitted group, fit the symmetric block matrix by the exact Bernoulli working criterion, and perform at least one exact Bernoulli profile-LRT assignment using R°. One pass suffices. The same theorem permits a finite number of the stated hard refits, including adaptive stopping, by the invariant-basin argument proved below.

Both algorithms return a fixed partition, or their initial partition, if a prescribed operation is undefined. This includes zero degrees, insufficient retained rank, a hollow denominator zero, an empty initial group, or an inadmissible parameter fit. The required operations are all well-defined on the good event, so this convention costs only its actual failure probability.

Theorem 1.1 (Ordinary SBM, literal constant oracle-risk ratio)**.** *Fix $K \geq 2$, $\pi_- > 0$ and $\kappa > 0$. Suppose*

$$n_a \geq \pi_- n, \qquad p/2 \leq P_{ab} \leq p \leq 1/4, \qquad \sigma_{\min}(P) \geq \kappa p, \qquad d = np \geq 5 \log n.$$

Use (8) and the finite growing/replacement likelihood decoder, with probability floor $\varepsilon = c_0 \max(1, n^{-1} \sum_i D_i)/n$ for a fixed sufficiently small $c_0 > 0$ (for example any $c_0 < 1/16$ on this class). There are constants C, n_0, depending only on the displayed fixed structural constants and c_0, such that for $n \geq n_0$,

$$\mathbb{E}\ell(\widehat{z}_E, z) \leq C R_{\mathrm{SBM},*}(n). \tag{11}$$

The algorithm uses neither P, p, nor the true partition. This includes fixed arbitrary K, unbalanced group sizes of comparable order, and general symmetric full-rank block probability matrices in the displayed range. It does not require all within-group probabilities, or all between-group probabilities, to be identical.

Theorem 1.2 (General bounded-parameter DC-SBM versus the profile oracle)**.** *Assume (2)–(3). Fix $C_1 < \infty$ and suppose*

$$C_0 \log n \leq d = np, \qquad np^2 \leq C_1, \tag{12}$$

where C_0 is sufficiently large depending only on the fixed structural bounds and C_1. The degree-metric decoder (9)–(10), with the deterministic initialization and estimated-parameter profile procedure just specified, satisfies

$$\mathbb{E}\ell(\widehat{z}_D, z) \leq C R_{\mathrm{prof},*}(n) \tag{13}$$

for all sufficiently large n, with C independent of n. The statement is uniform over the displayed compact parameter class. No correct grouping, true degree parameters, true block matrix, sample splitting, or leave-one-out eigensystem is an input to the algorithm.

Corollary 1.3 (The known-target-degree oracle on equal exposures)**.** *Under theorem 1.2, suppose additionally that the actual finite-n block exposures satisfy*

$$\sum_b (n_b/n) C_{ab} = S \quad \text{for all } a. \tag{14}$$

Then

$$R_{\mathrm{prof},*} \asymp R_{\theta,*}, \qquad \mathbb{E}\ell(\widehat{z}_D, z) \leq C R_{\theta,*}.$$

Thus there are nontrivial fixed-K SBM and DC-SBM classes on which the spectral-only fitted likelihood procedure attains a constant multiple of the original known-parameter local LRT risk. The ordinary free-profile decoder, with a sufficiently small floor coefficient for its structural parameter class, also satisfies (11) for general positive full-rank C under (12) and (14), without the entry-ratio-two restriction of theorem 1.1.

Exact equality in (14) can be relaxed to exposure differences whose contribution to all likelihood contrasts is uniformly $O(1)$. For example fixed $C\pi = S\mathbf{1}$ with $n_a = n\pi_a + O(1)$ suffices. Equality of limiting exposures with no rate is not by itself sufficient. The general known-θ_i claim without an additional restriction is false: theorem 7.1 gives an explicit diverging risk ratio for every full-graph algorithm.

Remark 1.4 (What is and is not constant). The factor may depend on signal conditioning, group proportions, the allowed degree-parameter range and C_1. It is independent of n on the stated class. We do not assert a small numerical constant, a limit of the risk ratio, or uniformity when $\sigma_{\min}(C) \to 0$. The latter weak-signal extension needs separate quantitative estimates.

2 A sharp Bernoulli likelihood transfer

We give the probability step explicitly to distinguish a risk factor from an exponential-rate factor. Sharp Chernoff comparisons in community detection are also developed by Zhou and Li [6]. The argument here treats an independent training-dependent perturbation and directional profile scores directly.

Lemma 2.1 (Bounded tilted error preserves the sharp prefactor)**.** *Let X_j be independent Bernoulli variables with probabilities p_j, and let*

$$T = b_0 + \sum_j w_j X_j, \qquad K(s) = \log \mathbb{E} e^{-sT}.$$

Suppose $|w_j| \le W$, the minimizer s_* satisfies $\tau \le s_* \le 1 - \tau$ for a fixed $\tau > 0$, and under the probability measure Q proportional to $e^{-s_* T}$,

$$\mathbb{E}_Q T = 0, \quad vd \le \mathrm{Var}_Q(T) \le V_0 d, \quad \sum_j r_j \le V_0 d, \tag{15}$$

where $r_j = Q(X_j = 1)$. Put $I = -K(s_*)$. For every fixed $L < \infty$ and all sufficiently large d, uniformly over $|x| \le L$,

$$c_L d^{-1/2} e^{-I} \le \mathbb{P}(T < x) \le \mathbb{P}(T \le x) \le C_L d^{-1/2} e^{-I}. \tag{16}$$

Let $\mathcal{W}$ be independent of $(X_j)_j$ and let

$$\widehat{T} = T + b + \sum_j e_j X_j$$

have coefficients measurable with respect to $\mathcal{W}$. On a measurable training event suppose

$$\max_j |e_j| \le M, \qquad \left| b + \sum_j r_j e_j \right| \le B, \qquad \sum_j r_j e_j^2 \le V. \tag{17}$$

For fixed M, V and every real x,

$$\mathbb{P}(\widehat{T} \le x \mid \mathcal{W}) \le C \exp\{s_* x + B + e^M V/2\}\, d^{-1/2} e^{-I}. \tag{18}$$

The constant is uniform over the training event. In particular B can be a nonnegative training-measurable random bound, provided its indicated exponential moment is integrated when removing the conditioning.

Proof. Under Q, the coordinates remain independent Bernoulli variables,

$$r_j = \frac{p_j e^{-s_* w_j}}{1 - p_j + p_j e^{-s_* w_j}}.$$

The sum of their centered weighted third absolute moments is at most $2W^3 \sum_j r_j = O(d)$. Berry–Esseen and (15) therefore show that every interval of length one has Q probability at most $C/\sqrt{d}$, uniformly in its location. Since $d\mathbb{P}/dQ = e^{-I} e^{s_* T}$, partitioning $(-\infty, x]$ into unit intervals gives the upper bound in (16).

For the lower bound choose a fixed interval $(x - L_0, x)$, with L_0 sufficiently large. Its Gaussian probability under a centered normal variable of variance in $[vd, V_0 d]$ is at least $cL_0/\sqrt{d}$ for large d, uniformly over bounded x. Choose L_0 so this exceeds twice the Berry–Esseen error. The Q probability is then at least $c'/\sqrt{d}$, and $e^{s_* T}$ is bounded below by a positive constant on the interval. Using strict endpoints also handles arbitrary lattice atoms and tie rules.

For the perturbation, independence of training and the row allows us to regard b, e as fixed. The elementary Bernoulli exponential-moment bound gives

$$\begin{aligned}\log \mathbb{E} e^{-s_* \widehat{T}} &= -I - s_* b + \log \mathbb{E}_Q e^{-s_* \sum_j e_j X_j} \\ &\le -I - s_* \left(b + \sum_j r_j e_j \right) + \frac{s_*^2 e^{s_* M}}{2} \sum_j r_j e_j^2 \\ &\le -I + B + e^M V/2.\end{aligned}$$

Now tilt the original law by $e^{-s_* \widehat{T}}$. Under this new measure $\widehat{Q}$ the coordinates remain independent and their odds are those under Q multiplied by $e^{-s_* e_j}$. Consequently their probabilities and variances are within fixed factors of those under Q. Furthermore,

$$\left(\sum_j r_j (1 - r_j)(w_j + e_j)^2 \right)^{1/2} \ge \sqrt{vd} - \sqrt{V}.$$

For large d, the variance of $\widehat{T}$ under $\widehat{Q}$ is therefore between positive constant multiples of d. Its centered third absolute moments sum to $O(d)$ because $w_j + e_j$ are uniformly bounded and $\sum_j \widehat{r}_j = O(d)$. Berry–Esseen bounds every unit interval under $\widehat{Q}$ by $C/\sqrt{d}$, irrespective of its mean. The same interval summation, with the displayed exponential moment, proves (18). □

2.1 Application to the actual Bernoulli SBM oracle

For a true label a and a competitor b, the oracle contrast is

$$T_{i,ab} = \ell_{i,*}(a) - \ell_{i,*}(b) = b_{i,ab} + \sum_{j \ne i} w_{z_j}^{ab} A_{ij},$$

where

$$w_c^{ab} = \log \frac{P_{ac}(1 - P_{bc})}{P_{bc}(1 - P_{ac})}, \qquad b_{i,ab} = \sum_{j \ne i} \log \frac{1 - P_{az_j}}{1 - P_{bz_j}}. \tag{19}$$

Its exact Chernoff information is

$$I_{i,ab} = - \min_{0 \le s \le 1} \sum_{j \ne i} \log \left[P_{az_j}^{1-s} P_{bz_j}^{s} + (1 - P_{az_j})^{1-s} (1 - P_{bz_j})^{s} \right]. \tag{20}$$

On the compact positive full-rank classes used here, probabilities are comparable to p and bounded away from one, the weights are bounded, s_* stays away from the endpoints, and the tilted variance is comparable to d. To verify the last assertions, the derivatives of the transform at its endpoints are respectively minus and plus the two directional Bernoulli KL divergences, both of order d. Its second derivative is bounded by Cd. The row separation implied by $\sigma_{\min}(P) \ge \kappa p$ gives a lower bound cd on that derivative throughout $[0, 1]$: tilted probabilities are comparable to p and the sum of squared log-odds differences is at least cn. Thus lemma 2.1 applies uniformly to every pair.

The same conclusion holds for the profile contrasts defined below, with their own directional information $J_{i,a \to b}$. It is not necessary for those two directional exponents to coincide.

2.2 Fixed-K comparison and label alignment

Let q_i be the error probability of the specified oracle at vertex i. For any coherent family of its candidate scores, strict pairwise loss to a competitor implies an oracle error; an oracle error implies non-strict loss to some competitor. Hence

$$\max_{b \ne a} \mathbb{P}(T_{i,ab} < 0) \le q_i \le \sum_{b \ne a} \mathbb{P}(T_{i,ab} \le 0).$$

By (16),

$$q_i \asymp \frac{1}{\sqrt{d}} \sum_{b \neq z_i} e^{-I_{i,ab}} \tag{21}$$

for the SBM oracle; replace I by J for the profile oracle. The constants may include a factor depending on fixed K. Fitted partitions are aligned with the true groups by a single common permutation on the good event. Uniform score comparisons under that alignment imply

$$\mathbb{E}[\ell(\widehat{z}, z)\mathbf{1}\{\mathcal{G}\}] \leq \frac{1}{n} \sum_i \mathbb{P}(\text{aligned fitted error at } i, \mathcal{G}).$$

The ensuing pairwise bounds apply to fixed true coefficient vectors; a data-dependent matching does not become an independence assumption. Finally,

$$\mathbb{E}\ell(\widehat{z}, z) \leq CR_* + \mathbb{P}(\mathcal{G}^c). \tag{22}$$

Section 6 proves $\mathbb{P}(\mathcal{G}^c) = O(R_*)$ under the stated model conditions. It is this comparison, rather than merely $\mathbb{P}(\mathcal{G}^c) = o(1)$, that completes each theorem.

3 Actual normalized spectral geometry and leave-one-out control

This section proves the spectral premises used by both fitted decoders. Leave-one-out (LOO) matrices occur only in the proof. The implemented encoder retains one full-graph eigensystem.

3.1 Raw concentration and deterministic spectral consequences

For any prescribed fixed $H > 0$, when $d \geq c \log n$ there is a fixed L_H such that

$$\mathbb{P}\{\|A - \Omega\|_{\text{op}} > L_H \sqrt{d}\} \leq e^{-Hd}. \tag{23}$$

Indeed, apply Corollary 3.12 and its centered-entry extension in Remark 3.13 of Bandeira–van Handel [2] to $A - \mathbb{E}A$. For bounded centered entries that result bounds the tail at $C\sqrt{\max_i \sum_j \operatorname{Var}(A_{ij})} + t$ by $n \exp(-t^2/C')$. Here the row variance sum is $O(d)$, and the entries have absolute value at most one. Choosing t a sufficiently large fixed multiple of $\sqrt{d}$ absorbs both H and $\log n$. The completed-mean diagonal costs $O(p)$. The centered-entry extension is needed: centered Bernoulli noise does not have a symmetric distribution.

Fix constants $a, b, L > 0$ and define the base event

$$\mathcal{E}_0 = \{\|A - \Omega\|_{\text{op}} \leq L\sqrt{d}, \quad ad \leq D_i \leq bd \text{ for every } i\}. \tag{24}$$

Its minimum-degree probability is deliberately not included in (23); it is treated separately in section 6. On $\mathcal{E}_0$,

$$\left\|(D - \bar{D})\mathbf{1}\right\|_2 \leq L\sqrt{nd}, \qquad \sum_j \frac{(D_j - \bar{D}_j)^2}{D_j} \leq Cn, \quad \bar{D} = \operatorname{diag}(\Omega\mathbf{1}). \tag{25}$$

These follow by applying $A - \Omega$ to $\mathbf{1}$; no uniform vanishing relative degree error is required.

The observed-degree-normalized completed mean

$$M = D^{-1/2}\Omega D^{-1/2}$$

has rank K, column space $\operatorname{col}(D^{-1/2}\Theta Z)$, and its nonzero eigenvalues have absolute values at least a fixed positive constant. For the last statement, $Z^\top \Theta D^{-1} \Theta Z$ is diagonal with entries comparable to n/d, and the nonzero eigenvalues are those of

$$p(Z^\top \Theta D^{-1} \Theta Z)^{1/2} C (Z^\top \Theta D^{-1} \Theta Z)^{1/2}.$$

The lower singular-value assumption supplies the fixed gap. Since $\|S - M\|_{\mathrm{op}} \le C/\sqrt{d}$, the retained eigenvalues are bounded away from zero, the discarded spectrum is $O(d^{-1/2})$, and

$$\left\|UU^\top - \Pi_M\right\|_{\mathrm{op}} \le C/\sqrt{d}, \tag{26}$$

where Π_M projects onto $\mathrm{col}(D^{-1/2}\Theta Z)$. These gap estimates follow directly by expanding a retained vector in the signal and complementary eigenbases, or by the usual subspace perturbation inequality. Positive and negative signal eigenvalues are both retained.

The same conclusions hold for every vertex-deleted graph with its degrees *recomputed*: deleting a vertex preserves the raw principal-submatrix norm bound, reduces each surviving degree by at most one, and changes a block mass by only a bounded amount. All constants can be chosen common for the full graph and these n minors.

3.2 Exact generalized equations for one incident row

Put the tested vertex first and write

$$A = \begin{pmatrix} 0 & x^\top \\ x & B \end{pmatrix}, \qquad k = \|x\|_1, \quad \Delta = \mathrm{diag}(B\mathbf{1}), \quad D = \mathrm{diag}(k, \Delta + E_x), \quad E_x = \mathrm{diag}(x). \tag{27}$$

Here $x \in \{0,1\}^{n-1}$. Suppose the deleted degrees lie between fixed multiples of d, $0 < k \le bd$, and

$$S_B = \Delta^{-1/2} B \Delta^{-1/2} = U_B \Lambda_B U_B^\top + N, \quad U_B^\top U_B = I_K, \quad N U_B = 0,$$

with signal eigenvalues separated from zero and $\|N\|_{\mathrm{op}} \le C/\sqrt{d}$. Write

$$V = \Delta^{-1/2} U_B, \quad Q_\perp = I - U_B U_B^\top, \quad F = \Delta^{-1/2} E_x \Delta^{-1/2}, \quad h = \Delta^{-1/2} x.$$

Then $\|F\| \le C/d$ and $\|h\|^2 \le Ck/d$. The full normalized matrix differs from $\mathrm{diag}(0, S_B)$ by $O(d^{-1/2})$: its off-diagonal normalized row has squared norm at most C/d, and the lower degree-renormalization perturbation is $O(d^{-1})$. The full retained signal therefore has the same fixed gap.

Take full generalized eigenvectors $g_r = D^{-1/2} u_r$, so $A g_r = \mu_r D g_r$ and jointly $G^\top D G = I_K$. Decompose them exactly as

$$g_r = \begin{pmatrix} t_r \\ \Delta^{-1/2}(U_B y_r + z_r) \end{pmatrix}, \qquad U_B^\top z_r = 0.$$

Let $h_V = U_B^\top h$ and $h_Q = Q_\perp h$. The top row and complementary equations give

$$h^\top (U_B y_r + z_r) = \mu_r k t_r, \tag{28}$$

$$(\mu_r I - N + \mu_r Q_\perp F Q_\perp) z_r = t_r h_Q - \mu_r Q_\perp F U_B y_r. \tag{29}$$

The inverse T_r of the complementary operator has bounded norm. Substituting into (28) gives

$$(\mu_r k - h_Q^\top T_r h_Q) t_r = h_V^\top y_r - \mu_r h_Q^\top T_r Q_\perp F U_B y_r. \tag{30}$$

Its denominator has magnitude at least ck: the second term is at most Ck/d. This remains true when k is small.

Put $M_B = \|U_B\|_{2\to\infty}$. The coordinates y_r have bounded norm because $\Delta^{1/2} G_{-i} = (I + F)^{-1/2} U_{-i}$ has norm at most one. Using

$$\|h_V\| \le C k M_B/\sqrt{d}, \qquad \|F U_B\|_F \le C M_B \sqrt{k}/d$$

in the exact equations yields

$$|t_r| \le C M_B/\sqrt{d}, \qquad \|z_r\|_2 \le C M_B \sqrt{k}/d. \tag{31}$$

The matrix $Y = (y_1, \dots, y_K)$ has bounded inverse: the full and padded deleted signal spaces have angle $O(d^{-1/2})$, and the extra multiplier $(I + F)^{-1/2}$ costs only $O(d^{-1})$.

Lemma 3.1 (Normalized row bounds from model events)**.** *Under the structural model assumptions, for any prescribed $H > 0$ there are row-concentration events of total failure probability at most $2Kne^{-Hd}$ such that, on their intersection with $\mathcal{E}_0$, the full normalized retained vectors and every deleted normalized retained basis satisfy*

$$\|U\|_{2\to\infty} \le C_H/\sqrt{n}, \qquad \max_i \left\|U^{(-i)}\right\|_{2\to\infty} \le C_H/\sqrt{n}. \tag{32}$$

They also satisfy the gap and angle bounds above. In particular, with $\Pi_0 = Z(Z^\top Z)^{-1}Z^\top$,

$$\|\Pi_X - \Pi_0\|_{\mathrm{op}} \le C_H/\sqrt{d}, \quad \|Q\|_{2\to\infty} \le C_H/\sqrt{n}, \tag{33}$$

and the same holds for all recomputed deleted projectors.

Proof. Let $M_{\mathrm{full}} = \|U\|_{2\to\infty}$ and $M_i = \left\|U^{(-i)}\right\|_{2\to\infty}$. The lower rows of the full eigenbasis are $(I+F)^{1/2}(U_B Y + Z_\perp)$, with $Z_\perp = (z_1, \dots, z_K)$. Using Y^{-1} and (31) gives

$$M_i \le CM_{\mathrm{full}} + CM_i/\sqrt{d}, \qquad M_i \le C' M_{\mathrm{full}} \quad \text{for large } d.$$

This argument has not assumed row incoherence. For the tested full row, (30), $k \asymp d$ and $u_{i,r} = \sqrt{k}\, t_r$ imply the sharper estimate

$$\|U_{i\cdot}\|_2 \le \frac{C}{\sqrt{d}} \left\|x^\top \Delta^{-1/2} U_B\right\|_2 + CM_i/d. \tag{34}$$

Conditional on B, its eigenbasis and recomputed degrees are fixed and x is an independent Bernoulli row. Its mean action has norm at most $C\sqrt{d/n}$, by $\|\mathbb{E}x\|_2 \le Cp\sqrt{n}$ and $\left\|\Delta^{-1/2}U_B\right\| \le C/\sqrt{d}$. We use the following weighted Bernoulli bound from Abbe–Fan–Wang–Zhong [1, Lemma 7, Eq. (C.2)]: for fixed v and independent Bernoulli coordinates with success probabilities at most q,

$$\mathbb{P}\left\{\left|\sum_j v_j(X_j - \mathbb{E}X_j)\right| > \frac{(2+\alpha)nq\,\|v\|_\infty}{1 \vee \log(\sqrt{n}\,\|v\|_\infty / \|v\|_2)}\right\} \le 2e^{-\alpha nq}. \tag{35}$$

Apply it to each of the fixed K columns of $\Delta^{-1/2}U_B$, taking $q = \min(Cp, 1)$, which is comparable to p on the structural class, and selecting α to give rate Hd. The column norms are at most $C/\sqrt{d}$ and $CM_i/\sqrt{d}$ in ℓ_2 and ℓ_∞, respectively. The function $b/(1 \vee \log(\sqrt{n}b/s))$ is nondecreasing in both admissible norm arguments. Thus outside conditional probability $2Ke^{-Hd}$,

$$\left\|x^\top \Delta^{-1/2} U_B\right\|_2 \le C\sqrt{d/n} + \frac{C_H\sqrt{d}\, M_i}{1 \vee \log(c\sqrt{n}M_i)}.$$

A union over i, followed by (34), yields

$$M_{\mathrm{full}} \le C/\sqrt{n} + \frac{C'_H M_{\mathrm{full}}}{1 \vee \log(c'\sqrt{n}M_{\mathrm{full}})} + C' M_{\mathrm{full}}/d.$$

For large d, absorb the last term. If $\sqrt{n}M_{\mathrm{full}}$ were larger than a sufficiently large fixed constant, the logarithmic term could also be absorbed, giving a contradiction. This proves (32) and its deleted versions.

The conditioning is legitimate even though $\mathcal{E}_0$ depends on x: define the conditional row event for every deleted graph satisfying its measurable degree and gap premises, apply (35) there, and use $\mathcal{E}_0$ only for deterministic implications afterwards.

For (33), bounded-condition-number multiplication by $D^{-1/2}$ in (26) compares $\mathrm{col}(X)$ with $\mathrm{col}(D^{-1}\Theta Z)$ at angle $O(d^{-1/2})$. Multiplying block a of $D^{-1}\Theta Z$ by $dS_a/\sqrt{n_a}$ gives entries $(\bar{D}_i/D_i)/\sqrt{n_a}$ on that block. By (25), this basis differs from $Z\,\mathrm{diag}(n_a^{-1/2})$ in Frobenius norm by $O(d^{-1/2})$. This proves the angle to the block space. Finally, $Q = X(X^\top X)^{-1/2}$ has the stated row bound since the eigenvalues of $X^\top X = U^\top D^{-1}U$ are comparable to $1/d$. The argument applies to DC-SBM as well as SBM. □

3.3 A deterministic full-to-LOO score lemma

Lemma 3.2 (Both actual full-graph reconstructions have bounded LOO error)**.** *Assume the deleted degree, spectral gap and discarded-spectrum conditions following (27), as well as* $\|U_B\|_{2\to\infty} \le C/\sqrt{n}$ *and* $k \le bd$*. For every* $w \in \mathbb{R}^n$ *with* $\|w\|_\infty \le W$*, let*

$$\Pi_B = V(V^\top V)^{-1}V^\top, \qquad H_B = VV^\top\Delta, \qquad V = \Delta^{-1/2}U_B.$$

Uniformly over such incident rows and coefficient vectors,

$$(R_E w)_i = x^\top \Pi_B w_{-i} + O(1), \tag{36}$$

$$(R_D w)_i = x^\top H_B w_{-i} + O(1). \tag{37}$$

The constants depend only on the displayed fixed bounds and W*. For the hollow degree-metric reconstruction on the model good event, (37) also holds with* R° *in place of* R_D*. The right-side coefficients use degrees recomputed from* B*. When* w *is fixed, as in the oracle contrasts, or is measurable from* B*, those coefficients are independent of the tested row. Uniform scalar stability alone does not make a full-row-dependent* w *independent.*

Proof. The case $k = 0$ has zero row action and can be given that convention. For $k > 0$, (31) now yields

$$\|t\| \le \frac{C}{\sqrt{nd}}, \qquad \left\|\Delta^{-1/2}Z_\perp\right\|_{\mathrm{op}} \le \frac{C\sqrt{k}}{d^{3/2}\sqrt{n}}. \tag{38}$$

For the Euclidean action right-multiply G by $Y^{-1}(V^\top V)^{-1/2}$. This does not change its column space and gives a basis

$$\mathcal{F} = \begin{pmatrix} \alpha \\ Q_B + E \end{pmatrix}, \quad Q_B = V(V^\top V)^{-1/2}, \quad \|\alpha\| \le C/\sqrt{n}, \quad \|E\| \le C\sqrt{k}/(d\sqrt{n}).$$

Its Gram matrix is $I + J$, where $\|J\| \le C\{\sqrt{k}/(d\sqrt{n}) + 1/n\}$. The deleted Euclidean basis has row norm at most $C/\sqrt{n}$, so

$$\left\|x^\top Q_B\right\| \le Ck/\sqrt{n}, \qquad \left\|x^\top E\right\| \le Ck/(d\sqrt{n}),$$
$$\left\|Q_B^\top w_{-i}\right\| \le W\sqrt{n}, \qquad \left\|\mathcal{F}^\top w - Q_B^\top w_{-i}\right\| \le C_W(1/\sqrt{n} + \sqrt{k}/d).$$

Expand

$$(R_E w)_i = (x^\top Q_B + x^\top E)(I + J)^{-1}\mathcal{F}^\top w$$

about $x^\top Q_B Q_B^\top w_{-i}$. The errors are bounded by a constant times

$$k/d + k/n + k\sqrt{k}/(d\sqrt{n}) = O(1), \tag{39}$$

because $k \le bd$ and $d \le Cn$. This proves (36). A coarse $O(n^{-1/2})$ bound for the lower rows alone would not suffice; the stronger E bound is essential.

For the degree-metric action, put $v_r = \Delta^{-1/2}z_r$, $f_r = Vy_r + v_r$ and $w_V = V^\top\Delta w_{-i}$. The exact top equation gives $k\mu_r t_r = x^\top f_r$, whence

$$(R_D w)_i = \sum_r (x^\top V y_r + x^\top v_r)\{kt_r w_i + y_r^\top w_V + v_r^\top \Delta w_{-i} + f_r^\top E_x w_{-i}\}. \tag{40}$$

The normalization $G^\top DG = I$ gives

$$Y^\top Y + Z_\perp^\top Z_\perp + ktt^\top + (VY + \Delta^{-1/2}Z_\perp)^\top E_x(VY + \Delta^{-1/2}Z_\perp) = I.$$

All terms except $Y^\top Y$ have norm at most $Ck/(nd)$, by (38) and the row bound on V. Since Y is square, $\left\|YY^\top - I\right\| \le Ck/(nd)$ as well. The leading term in (40) is $x^\top VYY^\top w_V$; its difference from $x^\top VV^\top \Delta w_{-i}$ is at most $C_W k^2/(nd)$. The other factors satisfy

$$\begin{aligned} \left\|x^\top V\right\| &\le Ck/\sqrt{nd}, & \|w_V\| &\le C_W\sqrt{nd}, \\ \left\|(x^\top v_r)_r\right\| &\le Ck/(d^{3/2}\sqrt{n}), & \left\|(v_r^\top \Delta w_{-i})_r\right\| &\le C_W\sqrt{k/d}, \\ \left\|(f_r^\top E_x w_{-i})_r\right\| &\le C_W k/\sqrt{nd}, & \|kt\| &\le Ck/\sqrt{nd}. \end{aligned}$$

Their products are bounded by constants times k/d, $k^2/(nd)$ and $k\sqrt{k}/(d\sqrt{n})$, all bounded. This proves (37) with negative as well as positive informative eigenvalues.

On the model good event, full row incoherence implies $|(H_D)_{ij}| \le C/n$, $|(R_D)_{ii}| \le Cd/n = Cp$ and $|s_i| \le Cp$. The hollow adjustment to a bounded row contrast is $s_i\{(H_D w)_i - w_i\} = O(p)$, proving the last assertion. □

The result compares the *actual* full-graph normalized algorithm to a row-independent score, with its changing degree normalizers included. It does not treat a full-graph eigenvector as independent of the row that created it. This is the dependence step needed before any conditional sharp-tail argument is applied.

4 Finite likelihood fitting and sharp risk in the ordinary SBM

We work under the ordinary-SBM assumptions of Theorem 1.1. In particular, $K \ge 2$ is fixed, $n_a \ge c_* n$, the block matrix is symmetric and uniformly full rank at scale p, and $p/2 \le P_{ab} \le p \le 1/4$. Write $d = np$, $C_a = \{i : z_i = a\}$, $M_0 = \Omega = ZPZ^\top$, and

$$\Pi_0 = Z(Z^\top Z)^{-1}Z^\top, \qquad p_{a*} = (P_{a,z_j})_{j=1}^n.$$

The diagonal of M_0 is retained in these comparison profiles; the observed graph has no self-loops. The resulting bounded diagonal correction is included below.

4.1 A finite decoder using only the retained information

Let $D = \operatorname{diag}(A\mathbf{1})$, and retain K normalized adjacency eigenpairs $D^{-1/2}AD^{-1/2}U = U\Lambda$, including the Perron direction and selecting the remaining signal eigenvalues by absolute value. On graphs where the encoder is defined, put

$$X = D^{-1/2}U, \qquad Q = X(X^\top X)^{-1/2}, \qquad Y = DX\Lambda(X^\top X)^{-1/2}, \qquad R = YQ^\top.$$

These are exact algebraic operations. The generalized eigenrelation gives $Y = AQ$ and hence $R = A\Pi$, where $\Pi = QQ^\top$. The Perron direction implies $\Pi\mathbf{1} = \mathbf{1}$ and $R\mathbf{1} = A\mathbf{1}$. In particular, the entire decoder below is evaluable from (D, U, Λ) without accessing an individual adjacency entry after compression. The exceptional-graph convention in the theorem makes the encoder and decoder total.

Fix $0 < c_0 < 1/16$, let $\bar{d} = n^{-1}\sum_i D_{ii}$, and set

$$\varepsilon = c_0 \max(1, \bar{d})/n, \qquad \mathcal{I} = [\varepsilon, 1-\varepsilon].$$

All clipping in this section is coordinatewise to $\mathcal{I}$. For $v \in \mathbb{R}^n$ and $\mu \in \mathcal{I}^n$, define

$$\ell(v;\mu) = \sum_j \{v_j \log \mu_j + (1-v_j)\log(1-\mu_j)\}, \qquad \ell_i(\mu) = \ell(R_{i*};\mu).$$

The entries of R may be signed. The score is nevertheless finite and has the retained-factor expression

$$\ell_i(\mu) = Y_i Q^\top \operatorname{logit}(\mu) + \sum_j \log(1-\mu_j). \tag{41}$$

For s fitted profiles $\mathcal{C} = (\mu_1, \dots, \mu_s)$ and simplex weights π, use

$$\mathcal{L}(\mathcal{C}, \pi) = \sum_i \log \sum_{a=1}^{s} \pi_a e^{\ell_i(\mu_a)}.$$

An E-step and its subsequent M-step are

$$\gamma_{ia} = \frac{\pi_a e^{\ell_i(\mu_a)}}{\sum_b \pi_b e^{\ell_i(\mu_b)}}, \qquad N_a = \sum_i \gamma_{ia},$$
$$\mu_{aj}^{\text{new}} = \operatorname{clip}_{\mathcal{I}}\left(\frac{\sum_i \gamma_{ia} R_{ij}}{N_a}\right), \qquad \pi_a^{\text{new}} = N_a/n. \tag{42}$$

When $N_a = 0$, the old profile is retained and the new weight is zero. The inner fitting subroutine performs at most 100 exact E/M updates. It may stop earlier when its labels are unchanged and its objective increment is at most $10^{-12}(1 + |\mathcal{L}_{\text{new}}|)$. Ties in all maximizations are settled deterministically. The proof uses only finiteness and monotonicity of these inner updates.

The candidate attached to vertex j is $\nu_j = \operatorname{clip}_{\mathcal{I}}(R_{j*})$. Start from the clipped global mean and fit one component. For $s = 2, \dots, K$, append a candidate maximizing

$$\sum_i \left(\ell_i(\nu_j) - \max_{a<s} \ell_i(\mu_a)\right)_+,$$

excluding the vertex identifiers selected in previous growing stages, and fit all s components from uniform weights. This gives K profiles. Next perform at most $T = \lceil \log n \rceil$ replacement rounds. For each deletion a from the round's incumbent, put

$$h_i^{(-a)} = \max_{b \neq a} \ell_i(\mu_b)$$

and choose a candidate maximizing $\sum_i (\ell_i(\nu_j) - h_i^{(-a)})_+$ over all vertices. Replace profile a by this candidate, initialize all K weights uniformly, and run the inner fit. After trying every deletion, retain the fit with largest $\mathcal{L}$, including the unchanged incumbent among the choices. A round without strict improvement terminates replacement.

Finally, evaluate the selected fit's responsibilities, perform one mandatory M-step in (42), and return

$$\widehat{z}_i \in \arg\max_a \{\ell_i(\mu_a^{\text{new}}) + \log \pi_a^{\text{new}}\}. \tag{43}$$

The reference to an E/M update here is ordered: the E-step is evaluated at the selected fit, its M-step follows, and the decoder uses the new profiles and weights.

4.2 A common box and deterministic reconstruction bounds

Throughout the deterministic arguments, intersect the base event $\mathcal{E}_0$ in (24) with the row-concentration events of Lemma 3.1. Its constants are fixed, independent of n. In addition to the degree bounds and normalized signal gap, its consequences used here are

$$\|A - M_0\|_{\text{op}} \le C\sqrt{d}, \qquad \|U\|_{2\to\infty} \le C/\sqrt{n}, \qquad \|\Pi - \Pi_0\|_{\text{op}} \le C/\sqrt{d}. \tag{44}$$

The same conclusions hold for the vertex-deleted graphs on their corresponding good events. Their probability costs are handled in the probability closure, after the sharp row calculation.

Lemma 4.1 (Reconstruction and generated probability box)**.** *On the full-graph good event, for all sufficiently large n,*

$$\|R - M_0\|_F^2 \le Cd, \qquad \max_{i,j} |R_{ij}| \le Cp, \qquad \|R_i(I - \Pi_0)\|_2 \le C\sqrt{p}, \tag{45}$$

$$\|\Pi_0(A - R)\Pi_0\|_{\mathrm{op}} \le C. \tag{46}$$

There are fixed $a_-, a_+, \eta_- > 0$ such that every profile generated by the decoder, every clipped true-class mean, and every true comparison profile belongs to

$$\mathcal{P}_n = \{\mu : \ a_- p \le \mu_j \le a_+ p, \quad \mu_j \le 1 - \eta_- \text{ for every } j\}. \tag{47}$$

Every true probability is at distance at least cp from both endpoints of the actual clipping interval $\mathcal{I}$.

Proof. Write $E = A - M_0$. Since $R = A\Pi$,

$$R - M_0 = E\Pi + M_0(\Pi - I).$$

Each term has rank at most K, operator norm at most $C\sqrt{d}$, and hence Frobenius norm at most $C\sqrt{d}$, by (44) and $\|M_0\|_{\mathrm{op}} \le Cd$. For the other row bounds, write $G = X^\top X$. The degree bounds imply that the eigenvalues of G are between constant multiples of d^{-1}. Thus $Q = XG^{-1/2}$ has row norms at most $C/\sqrt{n}$, and

$$AQ = DQT_0, \qquad T_0 = G^{1/2}\Lambda G^{-1/2}, \qquad \|T_0\|_{\mathrm{op}} \le C.$$

It follows that $\|(AQ)_i\|_2 \le Cd/\sqrt{n}$, which proves the entrywise bound for $R = (AQ)Q^\top$. Moreover,

$$\|R_i(I - \Pi_0)\|_2 \le \|(AQ)_i\|_2 \left\|Q^\top(I - \Pi_0)\right\|_{\mathrm{op}} \le C\sqrt{d/n}.$$

Both Π and Π_0 are orthogonal projectors. Consequently,

$$\|\Pi_0(I - \Pi)\Pi_0\|_{\mathrm{op}} = \|(I - \Pi)\Pi_0\|_{\mathrm{op}}^2 \le C/d.$$

Using $M_0 = \Pi_0 M_0 \Pi_0$ gives

$$\Pi_0(A - R)\Pi_0 = M_0\Pi_0(I - \Pi)\Pi_0 + \Pi_0 E(I - \Pi)\Pi_0.$$

The two terms have norms at most Cd/d and $C\sqrt{d}/\sqrt{d}$, respectively. This proves (46) without a commutation assumption on A and Π.

The average degree satisfies

$$\left|\bar{d} - n^{-1}\mathbf{1}^\top M_0 \mathbf{1}\right| \le \|E\|_{\mathrm{op}} \le C\sqrt{d}.$$

Since $d/2 \le n^{-1}\mathbf{1}^\top M_0 \mathbf{1} \le d$, eventually $d/4 \le \bar{d} \le 2d$. In particular $\bar{d} > 1$ eventually, so

$$c_0 p/4 \le \varepsilon \le 2c_0 p < p/8. \tag{48}$$

Together with $P_{ab} \in [p/2, p]$ and $p \le 1/4$, this gives the claimed interior distance from the clipping endpoints.

Each active raw M-step coordinate is a convex combination of a column of R, even though that column may have negative entries. It therefore lies in $[-C_R p, C_R p]$. Clipping makes it at most $\max(C_R p, \varepsilon) \le C_\mu p$. The initial mean and all clipped candidates obey the same bound; an inactive component retains an earlier profile. These observations prove inductively that all generated profiles lie between $c_0 p/4$ and $C_\mu p$. If $p \le 1/(2C_\mu)$, their upper bound is $1/2$. Otherwise the lower bound in (48) gives $\varepsilon \ge c_0/(8C_\mu)$, providing a fixed upper gap through clipping. Enlarge the constants to include the true profiles and the clipped true-class means. □

The box in (47) describes profiles generated by this algorithm and its comparison profiles. Uniform statements below concern this common box; they do not concern arbitrary vectors in the larger nominal interval $\mathcal{I}^n$.

Lemma 4.2 (Exact monotonicity for signed observations)**.** *Every exact update* (42) *increases or preserves* $\mathcal{L}$*.*

Proof. The log-sum-exp variational identity gives

$$\sum_{i,a} \gamma_{ia}\{\log \pi_a + \ell_i(\mu_a) - \log \gamma_{ia}\} \le \mathcal{L}(\mathcal{C}, \pi),$$

with equality at the E-step. For one profile coordinate and fixed responsibilities, put $a_j = \sum_i \gamma_{ia} R_{ij}$. Its objective is

$$a_j \log \mu + (N_a - a_j) \log(1 - \mu),$$

whose derivative has sign equal to $a_j/N_a - \mu$ when $N_a > 0$. Thus its global maximizer on $\mathcal{I}$ is the clipped weighted mean, whether or not a_j lies between zero and N_a. The weight maximizer is N_a/n. Zero mass leaves the corresponding coordinate objective constant. Maximization of this variational lower bound proves the claim. □

4.3 The finite replacement certificate and common alignment

For a K-profile list and weights, define the empirical reference loss

$$F_R(\mathcal{C}, \pi) = \sum_i \ell(R_i; p_{z_i,*}) - \mathcal{L}(\mathcal{C}, \pi),$$

and its population version

$$F_0(\mathcal{C}, \pi) = -\sum_i \log \sum_a \pi_a e^{-\mathrm{KL}(p_{z_i,*} \| \mu_a)}.$$

Here a vector Bernoulli divergence means the coordinate sum

$$\mathrm{KL}(p_{c*} \| \mu) = \sum_j \mathrm{KL}(\mathrm{Bern}(P_{c,z_j}) \| \mathrm{Bern}(\mu_j)).$$

The first term defining F_R is a comparison constant independent of the fitted list and weights. Hence every implemented likelihood comparison is precisely the reverse comparison of F_R. The population loss F_0 is nonnegative. Also put

$$G_R(\mathcal{C}) = \sum_i \{\ell(R_i; p_{z_i,*}) - \max_a \ell_i(\mu_a)\},$$
$$G_0(\mathcal{C}) = \sum_c n_c \min_a \mathrm{KL}(p_{c*} \| \mu_a).$$

These empirical reference losses are defined for signed R.

Lemma 4.3 (Uniform comparison of signed and population objectives)**.** *On the full-graph good event, uniformly over center lists in* $\mathcal{P}_n$ *and simplex weights,*

$$|F_R - F_0| + |G_R - G_0| \le Cn\sqrt{d}. \tag{49}$$

Always $G_R \le F_R$*. With uniform weights,* $F_R \le G_R + n \log K$*. Every true class supplies a candidate* ν_j *with* $\|\nu_j - p_{c*}\|_1 \le C\sqrt{d}$*; all these candidates are available in every replacement trial.*

Proof. For a true class c and fitted profile μ_a, the difference $\ell(v;\mu_a) - \ell(v;p_{c*})$ is affine in v, with coefficient $\mathrm{logit}(\mu_a) - \mathrm{logit}(p_{c*})$ bounded uniformly on $\mathcal{P}_n$. The maximum and log-sum-exp operations are 1-Lipschitz for changes bounded in maximum norm. Therefore each comparison in (49) is at most $C\sum_i \|R_i - p_{z_i,*}\|_1$. By (45), this sum is at most $n\,\|R - M_0\|_F \le Cn\sqrt{d}$. The common $\log p$ terms have canceled in the score contrasts used in this argument.

The inequalities between hard and soft reference losses follow from

$$K^{-1}e^{\max_a s_a} \le K^{-1}\sum_a e^{s_a} \le e^{\max_a s_a}$$

for uniform weights, and $\sum_a \pi_a e^{s_a} \le e^{\max_a s_a}$ for arbitrary weights. Finally, each class has at least $c_* n$ rows, so one of its rows has squared error at most Cd/n by the Frobenius bound. Its ℓ_1 error is at most $C\sqrt{d}$. Clipping toward an interval containing the true profile contracts coordinatewise absolute error. □

Lemma 4.4 (Finite replacement certificate)**.** *At the selected endpoint of the replacement stage,*

$$F_R \le \xi_n n d, \qquad \xi_n = C\{(1-1/K)^{\lceil \log n\rceil} + d^{-1/2}\} \longrightarrow 0. \tag{50}$$

The same conclusion holds if replacement stops because a complete round does not strictly improve the incumbent.

Proof. Assign each true profile a nearest fitted center in population divergence. A true class contributes at least G_0/K. If a fitted center is unused by this assignment, replacing that center by this true profile removes its contribution without increasing any other class's loss. If every center is used, the assignment between the K true classes and K fitted centers is a bijection; replacing the center of the highest-loss class has the same property. Consequently some ideal replacement has population hard loss at most $(1-1/K)G_0$.

The derivative of $\mathrm{KL}(p_{c*}\|\mu)$ in a parameter coordinate is $(\mu_j - P_{c,z_j})/[\mu_j(1-\mu_j)]$, uniformly bounded on the common box. Replacing the ideal profile by the available candidate from Lemma 4.3 costs at most $Cn\sqrt{d}$. Combining the objective comparisons gives a candidate list $\mathcal{C}'$ with

$$G_R(\mathcal{C}') \le (1-1/K)F_R(\mathcal{C},\pi) + Cn\sqrt{d}.$$

For a fixed deletion a, the empirical hard loss after inserting ν_j is exactly

$$\sum_i \{\ell(R_i;p_{z_i,*}) - h_i^{(-a)} - (\ell_i(\nu_j) - h_i^{(-a)})_+\}.$$

Thus the implemented maximum positive gain minimizes this loss among candidates. All deletions are tried. Uniform trial weights add at most $n\log K$ to the hard loss, and the subsequent exact inner fit cannot increase F_R by Lemma 4.2. Including the incumbent among the choices proves the recurrence

$$F_{R,t+1} \le (1-1/K)F_{R,t} + Cn\sqrt{d} + n\log K, \qquad F_{R,t+1} \le F_{R,t}. \tag{51}$$

Every generated profile lies in the common box, so each population divergence is at most Cd. Hence $F_{0,0} \le Cnd$ for arbitrary simplex weights, and the same bound holds for $F_{R,0}$ by (49). No preliminary accuracy guarantee for the growing stage is needed: its generated list supplies this finite starting bound. Iterating (51) for $\lceil \log n\rceil$ rounds gives (50), after enlarging its constant. If a complete round gives no strict improvement, its chosen loss is the incumbent's loss. The first inequality in (51) then immediately yields $F_R \le K(Cn\sqrt{d} + n\log K)$, which has the same conclusion. □

Lemma 4.5 (Common matching and refined responsibilities)**.** *The selected fit admits a single permutation after which*

$$\max_a \|\mu_a - p_{a*}\|_1 = o(d), \qquad -\log \min_a \pi_a = o(d). \tag{52}$$

Its next E-step satisfies

$$\frac{1}{n}\sum_{i,a} |\gamma_{ia} - Z_{ia}| \le C/d. \tag{53}$$

Proof. The preceding lemmas give $F_0 \le \zeta_n nd$ with $\zeta_n \to 0$. Since $F_0 \ge G_0$, each true class has a fitted center whose divergence is $o(d)$. On the common box, Bernoulli entropy curvature and Cauchy–Schwarz imply

$$\mathrm{KL}(p_{a*}\|\mu) \ge \frac{\|p_{a*} - \mu\|_2^2}{Cp} \ge \frac{\|p_{a*} - \mu\|_1^2}{Cd}.$$

Thus each true profile has a fitted center at $o(d)$ in ℓ_1. Uniform full rank and positive class sizes give

$$\min_{a\neq b} \|p_{a*} - p_{b*}\|_2^2 \ge cnp^2, \qquad \min_{a\neq b} \|p_{a*} - p_{b*}\|_1 \ge cd.$$

The nearby centers must therefore be distinct. There are exactly K of them, proving one common matching and the first assertion of (52).

For a row of true class a, each wrong-center population divergence is at least $c_1 d$ eventually, by this matching and the common-box Lipschitz bound. Its population soft loss is at least $-\log(\pi_a + e^{-c_1 d})$. It is the same for all n_a rows of that class, and every population soft loss is nonnegative. Consequently

$$\pi_a + e^{-c_1 d} \ge e^{-\zeta_n nd/n_a}, \qquad \pi_a \ge \tfrac{1}{2} e^{-C\zeta_n d}$$

for all sufficiently large n. This proves the weight assertion.

Fix a sufficiently small constant $t_0 > 0$. If $\|R_i - p_{a*}\|_1 \le t_0 d$ and $z_i = a$, the population separation, (52), and bounded contrast coefficients give a fitted correct-class margin at least $c_2 d$. The wrong responsibility mass on this row is at most $(K-1)e^{-c_2 d}$. Every remaining row has squared error at least $t_0^2 d^2/n$. The Frobenius bound limits their number to Cn/d. Summing their responsibility masses, which are at most one, proves (53). □

4.4 Fitted profiles change the relevant score by a bounded amount

Define the centered block fluctuation

$$\mathcal{B} = \max_{a,c} \left| \frac{\mathbf{1}_{C_a}^\top (A - \mathbb{E}A)\mathbf{1}_{C_c}}{\sqrt{n_a n_c}} \right|. \tag{54}$$

Lemma 4.6 (Block moments and deletion)**.** *For every fixed $t \ge 0$, $\mathbb{E}e^{t\mathcal{B}} \le C_t$. For a tested vertex i, let $\mathcal{B}_i$ be the analogous fluctuation calculated from $B = A_{-i,-i}$ and the deleted class sizes. Then $\mathbb{E}e^{t\mathcal{B}_i} \le C_t$, and on $D_{ii} \le Cd$,*

$$\mathcal{B} \le C(1 + \mathcal{B}_i). \tag{55}$$

Proof. Each normalized block sum in (54) is a sum of independent centered upper-triangular Bernoulli entries with coefficients of magnitude at most C/n. Its sum of coefficient-squared edge probabilities is at most Cp. The elementary centered Bernoulli transform bound therefore gives, for each fixed real t,

$$\log \mathbb{E} \exp\left\{ t \frac{\mathbf{1}_{C_a}^\top (A - \mathbb{E}A)\mathbf{1}_{C_c}}{\sqrt{n_a n_c}} \right\} \le Ct^2 e^{C|t|/n} p.$$

For example, this follows by applying $e^u - 1 - u \le u^2 e^{|u|}/2$ to every independent edge and multiplying their transforms. Taking both signs and summing over the fixed number of block pairs bounds $\mathbb{E}e^{t\mathcal{B}}$.

Deleting one vertex preserves the positive class-size bounds, so the same argument applies to $\mathcal{B}_i$. The unnormalized difference between a full and deleted block sum consists of at most twice the tested row's centered edge sum. Its absolute value is at most $C(D_{ii} + d)$. Dividing by $\sqrt{n_a n_c} \ge cn$ gives $O(p)$. The factor converting a deleted normalized block sum to the full denominator is bounded by a constant. This proves (55). □

For each class define the raw true-class mean and its clipped version

$$\bar{r}_a = n_a^{-1} \sum_{l \in C_a} R_l, \qquad \bar{\mu}_a = \mathrm{clip}_{\mathcal{I}}(\bar{r}_a).$$

These are comparison quantities only. For $a \neq b$, also define

$$w_j^{ab} = \log \frac{P_{a,z_j}(1 - P_{b,z_j})}{P_{b,z_j}(1 - P_{a,z_j})}, \qquad r_j^{ab} = \log \frac{1 - P_{a,z_j}}{1 - P_{b,z_j}}. \tag{56}$$

Lemma 4.7 (Uniform fitted-score comparison)**.** *On the full-graph good event, every M-step from responsibilities satisfying* (53)*, after their common matching, has weights bounded below and obeys*

$$\left| \ell_i(\mu_a) - \ell_i(\mu_b) + \log \frac{\pi_a}{\pi_b} - \left\{ R_i w^{ab} + \sum_j r_j^{ab} \right\} \right| \le C(1 + \mathcal{B}) \tag{57}$$

uniformly over i, a, b and all such responsibility matrices.

Proof. Write $e = n^{-1} \sum_{l,a} |\gamma_{la} - Z_{la}| \le C/d$. Then $N_a = n_a + O(ne)$, so $N_a \ge cn$ eventually and the updated weights N_a/n are bounded below. By (45), $\max_l \|R_l\|_1 \le Cd$. The difference of the raw responsibility-weighted mean and $\bar{r}_a$ can be written as a numerator perturbation and a denominator perturbation. Their combined ℓ_1 norm is at most

$$\frac{Cd}{N_a} \sum_l |\gamma_{la} - Z_{la}| + \frac{Cd}{N_a} |N_a - n_a| \le Ced \le C.$$

Clipping is 1-Lipschitz, whence

$$\|\mu_a - \bar{\mu}_a\|_1 \le C. \tag{58}$$

Jensen's inequality and the Frobenius bound give

$$\|\bar{r}_a - p_{a*}\|_2^2 \le n_a^{-1} \sum_{l \in C_a} \|R_l - p_{a*}\|_2^2 \le Cp.$$

Clipping toward an interval containing p_{a*} preserves this bound. The interior distance from Lemma 4.1 also gives the scalar inequality

$$|\,\mathrm{clip}_{\mathcal{I}}(x) - x| \le \frac{(x - u)^2}{cp}$$

for each true coordinate u. Thus $\|\bar{\mu}_a - \bar{r}_a\|_1 \le C$.

For a true block C_c, decompose the block sum of $\bar{r}_a - p_{a*}$ using $R - A$, $A - \mathbb{E}A$, and $\mathbb{E}A - M_0$. The first contributes at most

$$\sqrt{n_c/n_a}\ \|\Pi_0(R - A)\Pi_0\|_{\mathrm{op}} \le C,$$

by (46); the second is at most $C\mathcal{B}$. The diagonal correction contributes at most $P_{aa} \le p$. Including the clipping correction proves, for $\delta_a = \bar{\mu}_a - p_{a*}$,

$$\|\delta_a\|_2^2 \le Cp, \qquad \left|\sum_{j \in C_c} \delta_{aj}\right| \le C(1+\mathcal{B}). \tag{59}$$

For any profile in the common box, the score derivatives are

$$\frac{\partial \ell_i}{\partial \mu_j} = \frac{R_{ij} - \mu_j}{\mu_j(1-\mu_j)}, \qquad \frac{\partial^2 \ell_i}{\partial \mu_j^2} = -\frac{R_{ij}}{\mu_j^2} - \frac{1 - R_{ij}}{(1-\mu_j)^2}.$$

Their absolute values are at most C and C/p, respectively. Therefore (58) changes the score by $O(1)$, and the second-order remainder in a Taylor expansion at p_{a*} is at most $C\|\delta_a\|_2^2/p \le C$.

Split the first-order gradient at p_{a*} into its projection onto $\mathrm{col}(Z)$ and its orthogonal complement. Its block-constant projection has bounded coordinates; paired with δ_a, it is bounded by (59). The diagonal multiplier $[P_{a,z_j}(1 - P_{a,z_j})]^{-1}$ is block constant and commutes with Π_0. Hence the complementary gradient has norm at most

$$\frac{C}{p}\|R_i(I - \Pi_0)\|_2 \le C/\sqrt{p}.$$

Its inner product with δ_a, whose norm is at most $C\sqrt{p}$, is $O(1)$. We have proved $|\ell_i(\mu_a) - \ell_i(p_{a*})| \le C(1+\mathcal{B})$. Apply the same calculation to b and include the bounded updated log-weight ratio. The exact true-profile contrast is the expression in braces in (57), proving the result. □

4.5 Euclidean projected evidence at the oracle tilt

Fix a tested vertex i and labels $a \ne b$. Let $B = A_{-i,-i}$, $x = A_{-i,i}$, and let Π_B be the Euclidean projector onto the deleted generalized signal space from Lemma 3.2. Write $\Pi_{0,B}$ for the true block projector on the remaining vertices. For this paragraph, w and r denote the restrictions of (56) to these vertices. The exact oracle log likelihood ratio is

$$L_{i,ab} = b_{i,ab} + x^\top w, \qquad b_{i,ab} = \sum_{j \ne i} r_j^{ab}.$$

Let $I_{i,ab}$ be its exact Bernoulli Chernoff information in (20) and t_* its optimizing parameter, using the transform under class a with factor $e^{-t_* L_{i,ab}}$. Under the corresponding optimizing tilt, the row coordinates are independent with block-constant probabilities

$$q_j = \frac{P_{a,z_j}^{1-t_*} P_{b,z_j}^{t_*}}{P_{a,z_j}^{1-t_*} P_{b,z_j}^{t_*} + (1 - P_{a,z_j})^{1-t_*}(1 - P_{b,z_j})^{t_*}}. \tag{60}$$

These probabilities are comparable to p. The log-odds are bounded, and uniform full rank gives $\sum_{j \ne i} w_j^2 \asymp n$. For all tilt parameters in $[0,1]$, the log transform therefore has second derivative between cd and Cd. Its two endpoint values are zero. It follows that $I_{i,ab} \asymp d$, the optimizing parameter stays inside a fixed compact subinterval of $(0,1)$, and its tilted variance is comparable to d. Thus the oracle conditions of Lemma 2.1 hold uniformly.

Lemma 4.8 (Euclidean tilted cancellation)**.** *On the deleted-graph good event, let $\beta = (\Pi_B - I)w$. Then*

$$\max_j |\beta_j| \le C, \qquad |q^\top \beta| \le C, \qquad \sum_j q_j \beta_j^2 \le C. \tag{61}$$

Consequently, conditional on B, for every real u,

$$\mathbb{P}_a\{b_{i,ab} + x^\top \Pi_B w \le u \mid B\} \le C\frac{e^{-I_{i,ab} + t_* u}}{\sqrt{d}}. \tag{62}$$

Proof. The deleted Euclidean basis has row norms at most $C/\sqrt{n}$, so bounded entries of w imply bounded entries of $\Pi_B w$. Let $\delta = \|(I-\Pi_B)\Pi_{0,B}\|_{\mathrm{op}} \le C/\sqrt{d}$, as follows from (33) for the deleted graph. Both q and w belong to $\mathrm{col}(\Pi_{0,B})$. Orthogonality gives

$$q^\top \beta = -\langle (I-\Pi_B)q, (I-\Pi_B)w\rangle .$$

Since $\|q\|_2 \le Cp\sqrt{n}$ and $\|w\|_2 \le C\sqrt{n}$, its absolute value is at most $Cnp\delta^2 \le C$. Also $\|\beta\|_2^2 \le \delta^2 \|w\|_2^2 \le Cn/d$, so $\sum_j q_j\beta_j^2 \le Cp\|\beta\|_2^2 \le C$. This proves (61).

The conditional transform identity is

$$\mathbb{E}_a e^{-t_*(b_{i,ab}+x^\top \Pi_B w)} = e^{-I_{i,ab}} \mathbb{E}_q e^{-t_* x^\top \beta}.$$

Independence, $\log(1+y) \le y$, and $e^v - 1 - v \le v^2 e^{|v|}/2$ bound the final log transform by $C|q^\top\beta| + C\sum_j q_j\beta_j^2 \le C$. Under tilting by this estimated score, the row is still a product of Bernoulli variables. Its probabilities and complements are comparable to their oracle-tilted counterparts because the coefficient errors are bounded. Its score variance is of order d: in the oracle-tilted quadratic form,

$$\sum_j q_j(1-q_j)(w_j+\beta_j)^2 \ge \tfrac{1}{2}\sum_j q_j(1-q_j)w_j^2 - \sum_j q_j\beta_j^2 \ge cd$$

eventually, and the corresponding upper bound is Cd. The bounded summands have total third absolute centered moment $O(d)$. The bounds (61) give all three premises in (17), with zero additive intercept error and fixed constants. Lemma 2.1 now gives (62). A threshold u only contributes the displayed factor $e^{t_* u}$; it does not change this variance or the interval-concentration bound. □

Proposition 4.9 (Sharp error bound for the finite fitted decoder)**.** *Align the selected fit by the common matching from Lemma 4.5. Let $\mathcal{E}_i$ be the intersection of the full-graph and vertex-deleted good events, including the tested degree bound, used in Lemmas 3.1 and 3.2. For a vertex with true class a and any $b \ne a$, the final decoder (43) satisfies*

$$\mathbb{P}_z\{\widehat{z}_i \text{ is aligned to } b,\ \mathcal{E}_i\} \le C\frac{e^{-I_{i,ab}}}{\sqrt{d}}. \tag{63}$$

In particular, summing over competitors and averaging over vertices bounds the good-event expected misclassification loss by

$$\frac{C}{n}\sum_i \sum_{b\ne z_i} \frac{e^{-I_{i,z_i b}}}{\sqrt{d}}. \tag{64}$$

Proof. The finite replacement certificate and its mandatory final E/M update give the premise of Lemma 4.7. A wrong assignment from a to b, including an unfavorable score tie, implies on the full-graph good event that

$$R_i w^{ab} + \sum_j r_j^{ab} \le C(1+\mathcal{B}).$$

The full-to-deleted Euclidean identity (36) applies to the fixed bounded vector w^{ab}. Together with (55), it gives the deterministic event containment

$$\{\widehat{z}_i \text{ is aligned to } b,\ \mathcal{E}_i\} \subseteq \{b_{i,ab} + x^\top \Pi_B w_{-i}^{ab} \le C(1+\mathcal{B}_i),\ B \text{ good}\}. \tag{65}$$

The nonedge intercept changes by $r_i^{ab} = O(p)$ when the tested diagonal is removed, which is absorbed into the constant.

Use the full-graph event only for this deterministic containment. The right-hand event does not condition the tested row on full-graph eigenvectors, fitted parameters, or a degree event.

Conditional on B, the incident row has its original independent Bernoulli law. Applying (62) with $u = C(1 + \mathcal{B}_i)$, integrating over good B, and using Lemma 4.6 gives

$$\mathbb{P}_z\{\widehat{z}_i \text{ is aligned to } b,\ \mathcal{E}_i\} \le C\frac{e^{-I_{i,ab}}}{\sqrt{d}}\mathbb{E}e^{C\mathcal{B}_i} \le C'\frac{e^{-I_{i,ab}}}{\sqrt{d}}.$$

This proves (63). The common permutation is the same for every vertex, so the loss minimized over permutations is at most the loss under this matching. Summing the pairwise bounds proves (64). □

The sharp Bernoulli oracle comparison in Lemma 2.1 identifies the terms in (64) with local oracle errors up to fixed constants, including the $d^{-1/2}$ factor [6]. The probability closure pays the complement of these good events at that same oracle scale and completes Theorem 1.1.

Extension to general positive block profiles. The deterministic and conditional arguments also apply to a fixed positive full-rank block class

$$c_- p \le P_{ab} \le c_+ p, \qquad \max_{a,b} P_{ab} \le 1 - \epsilon_0,$$

with positive comparable class sizes and $d \to \infty$; p need not tend to zero. In particular $p \le p_{\max} := (1 - \epsilon_0)/c_-$. The average degree is eventually between $c_- d/2$ and $2c_+ d$. Choose

$$0 < c_0 < \min\{1/16,\ c_-/(4c_+),\ \epsilon_0/2\}.$$

Then $\varepsilon \le 2c_0 c_+ p < c_- p/2$, giving a distance at least $c_- p/2$ between every true probability and the lower clipping endpoint. For $n \ge 2$ the elementary graph bound $\bar{d} \le n - 1$ also gives $\varepsilon \le c_0 < \epsilon_0/2$. Hence the distance to the upper endpoint is at least $\epsilon_0/2 \ge \{\epsilon_0/(2p_{\max})\}p$. This proves the required interior condition for dense as well as sparse profiles.

The generated common-box proof, all likelihood derivatives, and all block-fluctuation moments now hold with fixed constants depending on c_-, c_+, ϵ_0. The exact oracle-tilted probabilities remain between the two candidate probabilities. Both these means and the exact log-odds vectors are block constant, so the Euclidean tilted calculation uses no sparse approximation. All arguments above therefore apply on the proved good event to this general class. Proposition 6.1 pays that event's complement under an explicit information–degree margin; Corollary 1.3 supplies the sparse equal-exposure case where the margin is automatic.

5 Degree-corrected SBM: the canonical spectral profile classifier

This section proves the degree-corrected result in theorem 1.2. The reconstruction used here is the projection in the observed-degree inner product. It is an explicit decoder specification: the preceding Euclidean reconstruction is not silently identified with it. All likelihoods below are Bernoulli likelihoods. The condition $np^2 \le C_1$ controls their approximation at a bounded *log-likelihood* scale.

Write $Z_{ia} = \mathbf{1}\{z_i = a\}$, $\Theta = \mathrm{diag}(\theta_1, \dots, \theta_n)$, $B = pC$, and

$$\Omega = \Theta Z B Z^T \Theta, \qquad D_0 = \mathrm{diag}(\Omega \mathbf{1}), \qquad D = \mathrm{diag}(D_1, \dots, D_n), \quad D_i = \sum_j A_{ij}.$$

The matrix Ω includes its diagonal; the observed graph is hollow. Thus $\|\Omega - \mathbb{E}A\|_{\mathrm{op}} = O(p)$. In particular, every raw noise bound used below is unchanged up to its fixed constant by this completion. Throughout this section a generic constant may depend on the fixed parameter bounds in theorem 1.2, but never on n. Constants depending also on a fixed initialization-basin radius M are denoted $C(M)$.

We use the common good event $\mathcal{G}$ supplied by lemma 3.1 and the probability calculation in section 6. Its consequences needed here are

$$cd \le D_i \le Cd, \qquad \|A-\Omega\|_{\mathrm{op}} \le C\sqrt{d}, \qquad \max_i \|U_{i\cdot}\|_2 \le Cn^{-1/2}, \tag{66}$$

$$\max_{a,b} \left|[Z^T(A-\Omega)Z]_{ab}\right| \le Cn, \tag{67}$$

together with the corresponding normalized spectral and degree consequences for the genuinely deleted graphs. The retained eigenvectors U are orthonormal, and their eigenvalues $\mu_1, \dots, \mu_K$ are the K largest in absolute value. The degree eigenvector is retained. The common lemmas establish these consequences from the model and pay their failure probabilities; they are not extra statistical assumptions. We now prove all remaining deterministic and local probabilistic steps.

5.1 The degree-metric reconstruction and its hollow correction

Put $S = D^{-1/2}AD^{-1/2}$, $\Pi = UU^T$, and

$$H = D^{-1/2}\Pi D^{1/2}, \qquad R = AH = D^{1/2}U\,\mathrm{diag}(\mu_1,\dots,\mu_K)U^TD^{1/2}. \tag{68}$$

The second formula evaluates R from retained spectral information. Although H need not be symmetric, $H^2 = H$ and $DH = H^TD$: it is an orthogonal projection for $\langle x, y\rangle_D = x^TDy$. Since the degree eigenvector is proportional to $D^{1/2}\mathbf{1}$, $H\mathbf{1} = \mathbf{1}$ and $R\mathbf{1} = D\mathbf{1}$.

For $h_i = R_{ii}$, define

$$s_i = \frac{h_i}{1-\Pi_{ii}}, \qquad R^\circ = R + \mathrm{diag}(s_1,\dots,s_n)(H-I). \tag{69}$$

On $\mathcal{G}$, the denominators are positive for all sufficiently large n. Direct substitution gives

$$R^\circ_{ii} = 0, \qquad R^\circ\mathbf{1} = D\mathbf{1}, \qquad R^\circ Q = AQ \quad \text{if } \mathrm{col}(Q) \subseteq \mathrm{col}(D^{-1/2}U). \tag{70}$$

The last identity follows because $HQ = Q$. The hollow correction need not be symmetric; a symmetric block parameter is fitted by the ordered Bernoulli criterion, equivalently by symmetrizing its pair contributions. The row profile scores always use the original rows of R°.

Lemma 5.1 (Entry and absolute-row bounds)**.** *On $\mathcal{G}$,*

$$\max_{i,j} |R^\circ_{ij}| \le Cp, \qquad \max_i \sum_j |R^\circ_{ij}| \le Cd, \qquad \max_i |s_i| \le Cp. \tag{71}$$

Every action in eqs. (68) and (69) is computable from (D, U, μ), with no access to individual edges after encoding.

Proof. The normalized adjacency has operator norm at most one. Consequently

$$|R_{ij}| \le \sqrt{D_iD_j}\,\|U_{i\cdot}\|_2\|U_{j\cdot}\|_2 \le Cd/n = Cp.$$

Moreover $|H_{ij}| \le \sqrt{D_j/D_i}\,\|U_{i\cdot}\|_2\|U_{j\cdot}\|_2 \le C/n$. These estimates give $|s_i| \le Cp$ and the entry bound for R°; summing its entries in absolute value gives the row bound. The displayed spectral formulas also prove the computational statement. □

5.2 A finite deterministic spectral initializer

The row-normalized population centers are particularly simple even when the θ_i's vary within blocks. To see this without replacing the observed degrees by their expectations, put

$$M_D = D^{-1/2}\Omega D^{-1/2}, \qquad G_D = Z^T\Theta D^{-1}\Theta Z.$$

The diagonal entries of G_D are between constant multiples of n/d. The K nonzero eigenvalues of M_D have absolute values bounded away from zero, by the lower singular-value bound on C. An orthonormal basis of its range has the form

$$V = D^{-1/2}\Theta Z G_D^{-1/2} O \tag{72}$$

for an orthogonal K-by-K matrix O. Each row has norm comparable to $n^{-1/2}$, and its normalization is exactly the z_i-th row of O. These K unit centers have pairwise distance $\sqrt{2}$.

By eq. (66), $\|S - M_D\|_{\mathrm{op}} \le C/\sqrt{d}$. The signal/bulk gap and Davis–Kahan therefore give an aligned basis with $\|U - V\|_F^2 \le C/d$. The elementary normalization inequality

$$\left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2 \le \frac{2\|u - v\|_2}{\|v\|_2}$$

holds when $u \neq 0$, and holds with a harmless change of constant under any fixed unit-vector convention for $u = 0$. For the normalized rows x_i and true unit centers c_a, it follows that

$$\sum_{i=1}^{n} \|x_i - c_{z_i}\|_2^2 \le Cn/d. \tag{73}$$

The following initializer requires no numerical K-means certificate. Starting with all rows remaining, repeat K times: choose a remaining row whose closed ball of radius $r = 1/8$ contains the largest number of remaining rows, break ties in a fixed order, record that row, and remove all remaining rows at distance at most $2r = 1/4$ from it. Finally assign every original row to its nearest recorded representative. If a step is undefined, use the algorithm's fixed fallback partition. The procedure uses $O(Kn^2)$ distance comparisons in fixed dimension.

Lemma 5.2 (Ball-count initialization)**.** *Suppose every true block contains at least γn rows and the unit centers are separated by $\sqrt{2}$. Put $\epsilon = 1/32$. If $\sum_i \|x_i - c_{z_i}\|_2^2 \le E$ and $m = E/\epsilon^2 < \gamma n/3$, the procedure above selects exactly one representative for each true center and, after one common permutation of labels, makes at most m errors. In particular, eq. (73) implies an initial error fraction at most C/d on $\mathcal{G}$.*

Proof. Call a row good if it is within ϵ of its true center. There are at most m bad rows. Inductively, all good rows of each previously selected block have been removed and all good rows of each unselected block remain. A radius-r ball centered at a good row of any unselected block contains every good row of that block, since $2\epsilon < r$. Its count is at least $\gamma n - m > m$. Hence the maximizing ball contains a good row from an unselected block. Its selected representative is within $r + \epsilon = 5/32$ of that block's center. Every good row of this block is within $r + 2\epsilon = 3/16 < 2r$ of the representative, while every good row of another block is at least $\sqrt{2} - r - 2\epsilon > 2r$ away. This proves the induction. After K selections, a good row is at most $3/16$ from its own representative and at least $\sqrt{2} - 3/16$ from every other one. Thus final nearest-representative assignment classifies all good rows correctly. □

5.3 Block-total accuracy and estimation of the nuisance parameters

We first record the residual estimate behind both the block fit and the tilted cancellation.

Lemma 5.3 (Block residual in the degree metric)**.** *On $\mathcal{G}$, with $F = (I - \Pi)D^{1/2}Z$,*

$$\|F\|_{\mathrm{op}} \le \|F\|_F \le C\sqrt{n}, \qquad \sum_i \frac{(D_i - D_{0,i})^2}{D_i} \le Cn. \tag{74}$$

The same conclusions hold for any genuinely deleted graph under its corresponding common good event.

Proof. The degree-vector bound follows from

$$\|(D - D_0)\mathbf{1}\|_2 = \|(A - \Omega)\mathbf{1}\|_2 \le C\sqrt{nd}$$

and $\min_i D_i \ge cd$. Write

$$D^{1/2}Z = D^{-1/2}D_0Z + D^{-1/2}(D - D_0)Z.$$

Since $D_{0,i} = p\theta_i \sum_c n_c C_{z_i c}$, the first term belongs to $\mathrm{col}(D^{-1/2}\Theta Z) = \mathrm{col}(M_D)$. Its Frobenius norm is $O(\sqrt{nd})$, while the angle between this space and $\mathrm{col}(U)$ is $O(d^{-1/2})$. Its residual is therefore $O(\sqrt{n})$. The squared Frobenius norm of the second term is exactly the degree sum in eq. (74), because every row of Z contains one unit entry. This proves the assertion. Deleting one vertex preserves all fixed group-mass bounds and the same calculation applies to the minor. □

Lemma 5.4 (Canonical reconstructed block totals)**.** *Fix $M < \infty$. Uniformly over every possibly data-dependent partition $\widehat{Z}$ with at most Mn/d mistakes after a common permutation,*

$$[\widehat{Z}^T R^\circ \widehat{Z}]_{ab} = n_a n_b p C_{ab} + O_M(n). \tag{75}$$

The usual diagonal exclusion is included in the $O_M(n)$ term.

Proof. Let $S_{\mathrm{tail}} = (I - \Pi)S(I - \Pi)$. Spectral invariance gives the exact quadratic identity

$$Z^T(R - A)Z = -F^T S_{\mathrm{tail}} F. \tag{76}$$

Because M_D has rank K, Weyl's inequality gives $\|S_{\mathrm{tail}}\|_{\mathrm{op}} \le C/\sqrt{d}$. By lemma 5.3, the right side of eq. (76) has norm at most $Cn/\sqrt{d}$. The hollow correction changes a true block total by at most Cd: indeed its total absolute contribution is at most

$$\max_i |s_i| \sum_i \sum_j |(H - I)_{ij}| \le Cp\, Cn = Cd.$$

Together with eq. (67) and $Z^T\Omega Z = \mathrm{diag}(n_1, \dots, n_K)(pC)\,\mathrm{diag}(n_1, \dots, n_K)$, this proves the claim for true labels.

Changing at most Mn/d labels changes at most CMn^2/d ordered pair memberships. Each reconstructed entry has size at most Cp, so the additional error is at most CMn. This deterministic argument is uniform over the entire basin and requires no independence of $\widehat{Z}$. □

For fitted labels $\widehat{z}$, let $\widehat{n}_a$ be their sizes, $\overline{D}_a^{\widehat{z}}$ their mean degrees, and set

$$\widehat{\theta}_j = \frac{D_j}{\overline{D}^{\widehat{z}}_{\widehat{z}_j}}. \tag{77}$$

Thus $\sum_{\widehat{z}_j = a} \widehat{\theta}_j = \widehat{n}_a$ exactly. Fit the symmetric matrix $\widehat{B}$ by maximizing

$$\frac{1}{2}\sum_{i \ne j} \left\{ R^\circ_{ij} \log(\widehat{\theta}_i\widehat{\theta}_j B_{\widehat{z}_i\widehat{z}_j}) + (1 - R^\circ_{ij}) \log(1 - \widehat{\theta}_i\widehat{\theta}_j B_{\widehat{z}_i\widehat{z}_j}) \right\} \tag{78}$$

over positive symmetric B, under a fixed probability cap below one. This is a precisely defined signed Bernoulli working criterion. The next lemma proves that its relevant roots exist and lie in the interior on $\mathcal{G}$.

Lemma 5.5 (Uniform fitted-parameter bounds)**.** *For every fixed M, uniformly over the basin in lemma 5.4, the degree estimates are bounded above and below by positive fixed constants and*

$$\frac{\|\widehat{\theta}-\theta\|_2}{\sqrt{n}} \le C(M)d^{-1/2}, \tag{79}$$

$$\max_{a,b}\left|\frac{\widehat{B}_{ab}}{pC_{ab}}-1\right| \le C(M)(d^{-1}+p) \le C(M)d^{-1}. \tag{80}$$

The last inequality uses $dp = np^2 \le C_1$.

Proof. Every fitted group has size comparable to n, and eq. (66) bounds its mean degree above and below by constant multiples of d. This proves boundedness of $\widehat{\theta}$. With true labels, write $D_i = dS_{z_i}\theta_i + \varepsilon_i$, where $S_a = \sum_c (n_c/n)C_{ac}$ and $\|\varepsilon\|_2 \le C\sqrt{nd}$. Division by the true-group mean degree, followed by subtraction of θ_i, gives normalized squared error $O(d^{-1})$. Changing an η-fraction of labels changes each fitted-group mean degree by at most $Cd\eta$. For correctly labeled vertices this changes the estimated theta by $O(\eta)$; incorrectly labeled vertices are an η-fraction and have bounded error. Hence

$$\frac{\|\widehat{\theta}-\theta\|_2}{\sqrt{n}} \le C(d^{-1/2}+\eta+\sqrt{\eta}),$$

which yields eq. (79) for $\eta \le M/d$.

For the block fit, group the ordered pairs in eq. (78) belonging to the unordered block pair $\{a,b\}$. Denote this ordered set by $\mathcal{J}_{ab}$, put $\nu_{ab} = 2 - \mathbf{1}\{a=b\}$, $t_{ij} = \widehat{\theta}_i\widehat{\theta}_j$, and $E^\circ_{ab} = \sum_{\mathcal{J}_{ab}} R^\circ_{ij}$. Its score equation is

$$E^\circ_{ab} = b\sum_{(i,j)\in\mathcal{J}_{ab}} \frac{(1-R^\circ_{ij})t_{ij}}{1-bt_{ij}}. \tag{81}$$

All $1-R^\circ_{ij}$ are positive for large n. The right side is strictly increasing in b below the cap. By lemma 5.4,

$$E^\circ_{ab} = \nu_{ab}n_an_bpC_{ab} + O_M(n) \asymp n^2p > 0.$$

For $b \asymp p$, bounded theta and $|R^\circ_{ij}| \le Cp$ give

$$\sum_{\mathcal{J}_{ab}} \frac{(1-R^\circ_{ij})t_{ij}}{1-bt_{ij}} = \sum_{\mathcal{J}_{ab}} t_{ij}\{1+O(p)\} = \nu_{ab}\widehat{n}_a\widehat{n}_b\{1+O(p)+O(n^{-1})\}.$$

The last equality uses exact fitted-block mean normalization; for $a=b$, removing the diagonal subtracts only $\sum_{\widehat{z}_i=a}\widehat{\theta}_i^2 = O(n)$. Evaluating eq. (81) at sufficiently small and large fixed multiples of p brackets its unique root, so

$$\widehat{B}_{ab} = \frac{E^\circ_{ab}}{\nu_{ab}\widehat{n}_a\widehat{n}_b}\{1+O(p)+O(n^{-1})\}.$$

Since $|\widehat{n}_a - n_a| \le Mn/d$, this proves eq. (80). The root is $O(p)$, hence satisfies the fixed cap strictly for large n. □

5.4 Exact Bernoulli profiling at bounded score error

Fix a tested vertex i with true block a. For candidate block b, define true neighbor masses and normalized profiles by

$$M_c^{(i)} = \sum_{\substack{j\ne i\\ z_j=c}} \theta_j, \qquad T_b^{(i)} = \sum_c M_c^{(i)}C_{bc}, \qquad q_{bc}^{(i)} = \frac{M_c^{(i)}C_{bc}}{T_b^{(i)}}. \tag{82}$$

The fitted analogues are

$$\widehat{M}_c^{(i)} = \sum_{\substack{j\neq i\\ \widehat{z}_j=c}} \widehat{\theta}_j = \widehat{n}_c - \widehat{\theta}_i \mathbf{1}\{\widehat{z}_i = c\}, \qquad \widehat{q}_{bc}^{(i)} = \frac{\widehat{M}_c^{(i)} \widehat{B}_{bc}}{\sum_e \widehat{M}_e^{(i)} \widehat{B}_{be}}. \tag{83}$$

The factor p is not needed to evaluate the fitted quantity. All these probabilities are bounded away from zero. By lemma 5.5 and fitted group-size accuracy,

$$\max_{i,a,b,c} \left| \log \frac{\widehat{q}_{ac}^{(i)}}{\widehat{q}_{bc}^{(i)}} - \log \frac{q_{ac}^{(i)}}{q_{bc}^{(i)}} \right| \leq C(M) d^{-1}. \tag{84}$$

Indeed normalization and logarithms are smooth on the fixed compact positive parameter set, and the relative mass error is $O_M(d^{-1} + n^{-1})$.

Lemma 5.6 (Bernoulli profile reduction)**.** *Let a hollow row $r = (r_j)_{j\neq i}$ have $\sum_j r_j = D_i \geq 0$, $\sum_j |r_j| \leq Ld$, and $D_i \leq Ld$. Suppose either $r_j \in \{0, 1\}$, or $\max_j |r_j| \leq Lp = o(1)$. For bounded positive neighbor parameters and block probabilities of order p, maximize the exact Bernoulli row criterion over the target parameter $t \geq 0$, with a fixed probability cap below one. After dropping terms independent of the candidate label, the profile contrast between a and b is*

$$L_{r,\mathrm{prof}}^{ab} = \sum_c N_c^r \log \frac{q_{ac}^{(i)}}{q_{bc}^{(i)}} + O(np^2), \qquad N_c^r = \sum_{\substack{j\neq i\\ z_j=c}} r_j. \tag{85}$$

The same assertion holds with fitted labels, thetas, block probabilities, and profiles. Fixed positive candidate prior probabilities add their bounded log-ratio to both sides.

Proof. For true parameters, omit the candidate-independent terms $\sum_j r_j \log \theta_j + D_i \log p$. The remaining score is

$$\ell_r(b, t) = D_i \log t + \sum_c N_c^r \log C_{bc} + \sum_{j\neq i} (1 - r_j) \log(1 - tp\theta_j C_{bz_j}).$$

For bounded t, expansion of the logarithm gives uniformly

$$\ell_r(b, t) = D_i \log t + \sum_c N_c^r \log C_{bc} - tpT_b^{(i)} + O(np^2). \tag{86}$$

The restored edge part of the linear exposure is bounded by $Cp \sum_j |r_j| \leq Cdp = Cnp^2$. The quadratic remainder is bounded by $Cp^2(n + \sum_j |r_j|) \leq Cnp^2$.

This expansion remains valid after maximizing. In both allowed row types $1 - r_j \geq 0$, so the exact score is concave in t. For a raw row, at least $n - O(d)$ nonedge terms remain; for a reconstructed row, every nonedge coefficient is at least $1/2$ for large n. Thus the nonedge part of its derivative has magnitude at least cd, whereas the positive term is D_i/t. The derivative is negative above a sufficiently large fixed t_0, uniformly in the candidate. The cap allows this bounded interval for large n, and the leading optimizer $t_b^0 = D_i/(pT_b^{(i)})$ is also bounded. Taking suprema in eq. (86) is therefore valid. The candidate-dependent leading maximum is $\sum_c N_c^r \log C_{bc} - D_i \log T_b^{(i)}$. Since $\sum_c N_c^r = D_i$, its difference is exactly the sum in eq. (85).

With fitted parameters replace pC by $\widehat{B}$ and the true masses by their fitted analogues. Boundedness and $\widehat{B} \asymp p$ give the identical remainder calculation. When $D_i = 0$, use the continuous extension after combining the terms into $D_i \log t$, with $0 \log 0 = 0$. Nonnegative nonedge coefficients make $t = 0$ optimal, and the same contrast identity holds. In particular a raw zero row gives zero contrast. □

Exact preservation of D_i is essential in this argument: it fixes the coefficient of $\log t$ and the profiled total-degree term. The proof needs no $O(d^{-1})$ estimate of an individual target theta.

Lemma 5.7 (Uniform replacement of all fitted stages)**.** *Fix M. For every partition in the M/d basin, fitted by eqs. (77) and (78), its actual row-profile contrast satisfies, simultaneously for all i, a, b,*

$$\widehat{L}_i^{ab} = [R^\circ w_{i,ab}]_i + O_M(1), \qquad (w_{i,ab})_j = \log \frac{q^{(i)}_{a,z_j}}{q^{(i)}_{b,z_j}}. \tag{87}$$

The formula is uniform over data-dependent labels and parameters obtained by these fits, under the same common label permutation.

Proof. The profile remainder is $O(np^2) = O(1)$. Replacing $\widehat{Z}$ by Z in a bounded block coefficient vector changes row i's score by at most

$$\max_j |R^\circ_{ij}|\, C\#\{j : \widehat{z}_j \neq z_j\} \le Cp\, CMn/d = C(M).$$

Replacing the fitted coefficients by their true counterparts costs at most $\sum_j |R^\circ_{ij}|\, C(M) d^{-1} \le C(M)$ by eq. (84). These are deterministic bounds holding for every partition in the basin, which proves the claimed uniformity. □

5.5 The profile reference and its sharp directional tails

For the fixed target i, true block a, and competitor b, write $w_j = (w_{i,ab})_j$, $v = \theta_i$, and

$$p_j = vp\theta_j C_{az_j}, \qquad T_{i,ab} = \sum_{j\neq i} w_j A_{ij}, \qquad K_{i,ab}(s) = \sum_{j\neq i} \log\{1 + p_j(e^{-sw_j} - 1)\}.$$

Set

$$J_{i,a\to b} = -\min_{s>0} K_{i,ab}(s). \tag{88}$$

This is the exact Bernoulli transform of the leading profiled contrast. It is not asserted to be the simple-hypothesis Chernoff information for the known-target-theta oracle.

Lemma 5.8 (Uniform profile saddle and reference scale)**.** *The minimizing $s = s_{i,ab}$ stays in a fixed compact subinterval of $(0, 1)$. Under the exact tilted product law,*

$$r_j = \frac{p_j e^{-sw_j}}{1 - p_j + p_j e^{-sw_j}}, \qquad \sum_j r_j w_j = 0, \qquad \sum_j r_j(1 - r_j) w_j^2 \asymp d. \tag{89}$$

Uniformly for bounded real thresholds x,

$$\mathbb{P}_a(T_{i,ab} < x) \asymp d^{-1/2} \exp\{-J_{i,a\to b}\}. \tag{90}$$

The exact full-edge Bernoulli profile-reference error at vertex i is comparable, by fixed constants, to

$$\sum_{b\neq a} d^{-1/2} \exp\{-J_{i,a\to b}\}. \tag{91}$$

Proof. Uniformly on bounded s intervals, the transform and its first two derivatives satisfy

$$d^{-1} K_{i,ab}(s) = v\, \frac{T^{(i)}_a}{n} \left\{ \sum_c (q^{(i)}_{ac})^{1-s} (q^{(i)}_{bc})^s - 1 \right\} + O(p).$$

The derivative of the leading expression is strictly negative at zero and strictly positive at one. Positivity, the lower singular-value bound on C, and the group-mass bounds make these signs

and the positive second derivative uniform. Indeed two normalized rows could coincide only if two rows of C were proportional; compactness of the stated parameter class excludes even an arbitrarily small normalized separation. Strict convexity now gives the interior saddle and tilted variance of order d.

The bounded-weight, variance, and third-moment hypotheses of lemma 2.1 hold, so its bounded-shift tail conclusion proves eq. (90). In particular, no nonlattice assumption is necessary. Alternatively, under the displayed tilt, Berry–Esseen bounds every unit-interval mass by $C/\sqrt{d}$; summing against e^{sT} gives the upper bound. A sufficiently long fixed interval immediately below x has Gaussian mass larger than twice the Berry–Esseen error, giving the lower bound.

The probability of $D_i > Ld$ can be made $\exp(-Ld\,c_L)$ with arbitrarily large fixed exponential rate by increasing the fixed upper cutoff. On its complement, lemma 5.6 compares the exact reference contrast to $T_{i,ab}$ up to a fixed additive constant. The upper-degree exception is negligible relative to the displayed tail scales. Each strict pairwise defeat of the true label implies an error, while an error is contained in the union of weak pairwise defeats. Fixed K, the two bounded-shift tail bounds, and fixed ties therefore prove eq. (91). □

5.6 The complete degree-weighted tilted cancellation

We now work on the genuinely deleted graph $A^{(-i)}$. To simplify notation within this subsection, D_-, U_-, Π_-, and

$$H_- = D_-^{-1/2}\Pi_- D_-^{1/2}$$

refer only to this independent training graph. Its completed mean is Ω_-, and $D_{0,-} = \mathrm{diag}(\Omega_- \mathbf{1})$. The tested row remains independent Bernoulli conditional on the deleted graph. Let $w = (w_j)_{j\neq i}$ be the true block coefficient vector above, and define $e = (H_- - I)w$.

Lemma 5.9 (Bounded tilted coefficient error)**.** *On the deleted-graph good event, the exact profile tilt in eq.* (89) *satisfies*

$$\max_j |e_j| \le C, \qquad \sum_j r_j e_j^2 \le C, \qquad \left|\sum_j r_j e_j\right| \le C. \tag{92}$$

All constants are uniform over the vertex and ordered pair of distinct candidate blocks.

Proof. The residual lemma on the deleted graph gives

$$\|(I - \Pi_-)D_-^{1/2} Z_-\|_{\mathrm{op}} \le C\sqrt{n}.$$

Consequently, for every bounded block vector w,

$$\|(I - H_-)w\|_{D_-} \le C\sqrt{n}, \qquad \|(I - H_-)w\|_2^2 \le Cn/d. \tag{93}$$

Incoherence bounds the coordinates of $H_- w$:

$$|(H_- w)_j| \le D_{-,j}^{-1/2}\|U_{-,j\cdot}\|_2 \|U_-^T D_-^{1/2} w\|_2 \le (C/\sqrt{d})(C/\sqrt{n})C\sqrt{nd} \le C.$$

This proves bounded coefficients. Since $r_j \le Cp$, eq. (93) gives $\sum_j r_j e_j^2 \le Cp\,Cn/d \le C$.

For the mean, exact Bernoulli tilting gives

$$r_j = vp\theta_j C_{az_j} e^{-sw_j} + \varepsilon_j, \qquad \max_j |\varepsilon_j| \le Cp^2. \tag{94}$$

For a remaining vertex in block c, write

$$T_c^- = \sum_{\ell\neq i} \theta_\ell C_{cz_\ell}, \qquad h_j = \frac{vC_{ac}e^{-sw_c}}{T_c^-}.$$

Then h is block constant with $\|h\|_\infty \le C/n$, and the leading vector in eq. (94) is exactly $D_{0,-}h$. Because H_- is D_--orthogonal,

$$r^T(H_- - I)w = -\langle (I-H_-)h, (I-H_-)w\rangle_{D_-} + \big((D_{0,-}-D_-)h\big)^T(H_- - I)w + \varepsilon^T(H_- - I)w. \tag{95}$$

This identity retains the random-degree correction explicitly. The first factor in its first term is at most $C/\sqrt{n}$: apply the block residual estimate to the block coefficients of h, which have magnitude $O(n^{-1})$. The second factor is at most $C\sqrt{n}$, so that term is $O(1)$.

For the next term, the degree aggregate in eq. (74) gives

$$\|(D_{0,-}-D_-)h\|^2_{D_-^{-1}} = \sum_j \frac{(D_{0,-,j}-D_{-,j})^2 h_j^2}{D_{-,j}} \le C/n.$$

Cauchy–Schwarz in the degree metric and eq. (93) again give $O(1)$. Finally, bounded e_j and eq. (94) bound the last term by $Cnp^2 = O(1)$. This proves the tilted mean bound and the lemma. □

The middle term of eq. (95) is why replacing random degrees by population degrees without a remainder is invalid. The product of two residual bounds is what reduces the leading tilted bias to a constant.

5.7 Completion of the risk bound on the common good event

By lemma 5.2, the initial partition lies in a fixed M/d basin on $\mathcal{G}$. Apply one nuisance fit and one profile assignment. The uniform comparison eq. (87) and the actual full-to-deleted-graph scalar lemma lemma 3.2 give

$$\widehat{L}_i^{ab} = \sum_{j\neq i} A_{ij}(H_- w_{i,ab,-i})_j + O_M(1) = T_{i,ab} + \sum_{j\neq i} A_{ij}e_j + O_M(1). \tag{96}$$

The leave-one-out graph appears only in this proof identity; the implemented classifier uses the single full-graph encoding. The $O_M(1)$ error is a deterministic uniform bound, so it need not be asserted to be independent of the tested row.

Condition on the genuinely deleted graph. On its measurable good event, lemma 5.9 supplies all three coefficient conditions of the sharp transfer lemma lemma 2.1. Therefore, for a fixed bounded threshold large enough to absorb eq. (96),

$$\mathbb{P}_a\left\{\sum_{j\neq i} A_{ij}(H_- w_{i,ab,-i})_j \le C(M) \,\middle|\, A^{(-i)}\right\} \le C(M)d^{-1/2}e^{-J_{i,a\to b}}. \tag{97}$$

The common full event implies the required minor event by the common geometry lemmas. Hence restricting an actual error to $\mathcal{G}$ is contained in the event on the left of eq. (97) together with the measurable minor event. This does not condition the tested row on a dependent full-graph event.

Integrating over the deleted graph and summing over $b \neq a$ gives

$$\mathbb{P}\{\widehat{z}_i \neq z_i, \mathcal{G}\} \le C\sum_{b\neq z_i} d^{-1/2}e^{-J_{i,z_i\to b}}.$$

Here labels are aligned by the common initial permutation. The permutation-invariant misclassification loss is no larger than the loss under that alignment. By lemma 5.8, the last sum is at most a constant times the exact full-edge profile-reference error at i. Averaging proves

$$\mathbb{E}\{\ell(\widehat{z}, z)\mathbf{1}\{\mathcal{G}\}\} \le CR_{\mathrm{prof},*}. \tag{98}$$

The calculation in section 6 proves $\mathbb{P}(\mathcal{G}^c) \le CR_{\mathrm{prof},*}$, including the minimum-degree exception at the correct exponential scale. Because the fallback loss is at most one, adding that contribution to eq. (98) completes theorem 1.2.

5.8 Optional repeated hard fitting and finite adaptive stopping

The same theorem holds if, after initialization, the algorithm repeatedly recomputes eqs. (77) and (78) and makes a hard profile assignment using the same R°. It may return any data-dependent finitely stopped iterate after at least one completed profile assignment. A finite iteration cap is one explicit termination rule.

Lemma 5.10 (A uniform hard-refit basin)**.** *There is a fixed M such that, on $\mathcal{G}$ and for all sufficiently large n, every hard fit-and-profile update maps the M/d error basin into itself, under the same label matching. Every error at every updated iterate is contained in*

$$[R^\circ w_{i,ab}]_i \le C(M) \quad \text{for some } b \ne z_i. \tag{99}$$

Proof. First,

$$\|R^\circ - \Omega\|_F^2 \le C_{\mathrm{spec}} d. \tag{100}$$

Indeed $S_K = U\,\mathrm{diag}(\mu)U^T$ is a best rank-K operator-norm approximation to S. Since M_D has rank K, $\|S_K - M_D\|_{\mathrm{op}} \le 2\|S - M_D\|_{\mathrm{op}} \le C/\sqrt{d}$. Their difference has rank at most $2K$, so its Frobenius norm is also $O(d^{-1/2})$. Multiplication by $D^{1/2}$ on both sides gives $\|R - \Omega\|_F \le C\sqrt{d}$. The hollow term has Frobenius norm at most

$$\max_i |s_i|\,\|H - I\|_F \le Cp\,C\sqrt{n} = Cd/\sqrt{n},$$

whose square is at most Cd, proving eq. (100).

The completed population contrast has a uniform positive margin: with $\lambda_i = \sum_{j\ne i} p\theta_i\theta_j C_{z_i z_j}$,

$$[\Omega w_{i,ab}]_i = \lambda_i\,\mathrm{KL}(q_a^{(i)} \| q_b^{(i)}) + O(p) \ge c_{\mathrm{gap}} d, \qquad a = z_i,\ b \ne a. \tag{101}$$

The $O(p)$ term is the completed diagonal. The uniform KL gap follows from the same compactness and full-rank argument as in lemma 5.8. Also $\|w_{i,ab}\|_2 \le C\sqrt{n}$.

For an input partition in a fixed M/d basin, lemma 5.7 bounds its fitted-contrast discrepancy by $C(M)$. For large n, this is at most $c_{\mathrm{gap}} d/2$. A new erroneous assignment therefore requires

$$\|(R^\circ - \Omega)_{i\cdot}\|_2 \ge \frac{c_{\mathrm{gap}} d}{2C\sqrt{n}}.$$

Summing this squared requirement and applying eq. (100) shows that the new iterate has at most $C_{\mathrm{new}} n/d$ errors. The constant C_{new} depends only on the fixed spectral and model constants, not on M. Choose M at least both C_{new} and the initialization constant. All sufficiently large n then give basin invariance. Group indices remain matched to the same true indices throughout: the uniform score comparison and its population margin use that fixed matching at every step.

Finally every erroneous assignment directly satisfies the fixed-threshold canonical event in the statement by lemma 5.7. The threshold is the same for all partitions in the basin. □

By induction, lemma 5.10 applies to every iterate on the single graph event $\mathcal{G}$. The final error of an adaptively chosen finite iterate is thus contained in the same canonical event used to prove eq. (97). Applying that probability bound once proves the same sharp risk bound; there is no union over iterations and no stopping-time independence assertion. This corollary concerns these specified hard updates. It does not assert the same result for an unrelated soft update or a reconstruction changed during iteration.

5.9 Equal exposures and the original known-target-theta oracle

We finish by proving corollary 1.3. Suppose

$$\sum_c \frac{n_c}{n} C_{ac} = S \quad \text{for every } a. \tag{102}$$

The block means of theta equal one, so $T_a^{(i)} = nS - \theta_i C_{a,z_i}$. In particular $T_a^{(i)} - T_b^{(i)} = O(1)$ and $\log(T_a^{(i)}/T_b^{(i)}) = O(n^{-1})$. Let $v = \theta_i$ be supplied to the full-edge known-target-theta oracle. Its exact Bernoulli contrast is

$$L_{A,\text{known}}^{ab} = \sum_c N_c \log \frac{C_{ac}}{C_{bc}} + \sum_{j \neq i} (1 - A_{ij}) \log \frac{1 - vp\theta_j C_{az_j}}{1 - vp\theta_j C_{bz_j}}.$$

On $D_i \leq Ld$, the same direct Bernoulli expansion used above gives

$$L_{A,\text{known}}^{ab} = \sum_c N_c \log \frac{C_{ac}}{C_{bc}} - vp(T_a^{(i)} - T_b^{(i)}) + O(np^2).$$

By lemma 5.6, the exact profile contrast is

$$L_{A,\text{prof}}^{ab} = \sum_c N_c \log \frac{C_{ac}}{C_{bc}} - D_i \log \frac{T_a^{(i)}}{T_b^{(i)}} + O(np^2).$$

Thus, using eq. (102),

$$|L_{A,\text{prof}}^{ab} - L_{A,\text{known}}^{ab}| \leq C(np^2 + p) = O(1). \tag{103}$$

Fixed prior log-ratios, if present, are identical on both sides. The bounded-shift upper and lower tails in eq. (90) show that the two directional error probabilities are comparable by constants. The upper-degree exception has arbitrarily large fixed exponential rate after enlarging L, and is negligible for both references. Finite unions over competitors and averaging over vertices yield

$$R_{\text{prof},*} \asymp R_{\theta,*}.$$

Combining this with theorem 1.2 proves corollary 1.3. The condition permits heterogeneous theta, unequal block sizes, and general positive full-rank symmetric C obeying its exposure constraints. Its role is precisely to remove the label information carried by the leading total-degree exposure, which profiling otherwise changes.

5.10 The classical balanced DC-SBM oracle and its sharp risk

The fundamental local testing problem used by Gao, Ma, Zhang and Zhou [3] fixes the target degree parameter under both candidate labels. Its exact simple-hypothesis likelihood test therefore corresponds to $R_{\theta,*}$ in section 1.2. The next proposition explicitly connects that reference, the profile reference and true-group neighbor voting in the balanced planted-partition model. This is an oracle comparison with its own assumptions: it uses neither the spectral good event $\mathcal{G}$ nor a logarithmic lower bound on np.

Proposition 5.11 (Sharp alignment of the classical DC-SBM references)**.** *Fix an arbitrary $K \geq 2$, positive bounds θ_-, θ_+, and $0 < r_- < r_+ < 1$. Let $n = Km$, let every group have m vertices, and suppose*

$$\theta_- \leq \theta_i \leq \theta_+, \qquad \sum_{i:z_i=a} \theta_i = m \quad \textit{for all } a.$$

Specialize the block matrix $B = pC$ to

$$B_{aa} = p, \qquad B_{ab} = q \ (a \neq b), \qquad r_- \leq q/p \leq r_+. \tag{104}$$

Assume only, for the present oracle comparison,

$$d = np \longrightarrow \infty, \qquad np^2 \le C_1 < \infty. \tag{105}$$

Define $R_{\mathrm{vote},*}$ as the average error of the local raw-row rule which knows the neighbor labels and selects the group with the largest incident block count. Use any fixed tie ordering. Then, uniformly over this parameter class,

$$\begin{aligned} R_{\theta,*} &\asymp R_{\mathrm{prof},*} \asymp R_{\mathrm{vote},*} \\ &\asymp \frac{1}{n}\sum_{i=1}^{n} \frac{1}{\sqrt{np}} \exp\left\{-\theta_i \frac{n}{K}(\sqrt{p}-\sqrt{q})^2\right\}. \end{aligned} \tag{106}$$

The profile reference is exactly the Bernoulli profile in (6), with candidate probabilities capped at $1/2$. The same comparison holds if the voting counts are normalized by the self-excluded neighbor-group sizes. Constants depend only on the displayed fixed bounds, not on n. In particular, (105) permits $np = o(\log n)$.

Proof. Fix a target i, its true group a, and a competitor $b \neq a$. For this proof write

$$N_c = \sum_{\substack{j\neq i \\ z_j=c}} A_{ij}, \qquad M_c = \sum_{\substack{j\neq i \\ z_j=c}} \theta_j = m - \theta_i \mathbf{1}\{c=a\}, \qquad L = \log(p/q).$$

Thus L is bounded above and below by positive constants. Put

$$V_{i,ab} = L(N_a - N_b), \qquad J_i = \theta_i m(\sqrt{p}-\sqrt{q})^2.$$

The assumptions imply $J_i \asymp d$, uniformly in i. They also imply $p \to 0$, so every true candidate edge probability is eventually less than $1/2$.

The exact Bernoulli count transform. Under the actual true-row law, independence gives

$$\begin{aligned} \Psi_{i,ab}(s) &:= \log \mathbb{E}_a e^{-sV_{i,ab}} \\ &= \sum_{\substack{j\neq i \\ z_j=a}} \log\{1+\theta_i\theta_j p(e^{-sL}-1)\} + \sum_{j:z_j=b} \log\{1+\theta_i\theta_j q(e^{sL}-1)\}. \end{aligned} \tag{107}$$

Taylor expansion, uniformly for $s \in [0,1]$, gives

$$\Psi_{i,ab}(s) = m\theta_i\{p(e^{-sL}-1) + q(e^{sL}-1)\} + O(np^2+p). \tag{108}$$

The same error bound holds for the first two derivatives. Bounded theta and bounded L make the sum of quadratic Bernoulli remainders $O(np^2)$. The additional $O(p)$ term accounts for the omitted θ_i in $M_a = m - \theta_i$.

The leading transform in (108) is minimized at $s = 1/2$, with minimum $-J_i$. Its endpoint derivatives have opposite signs, each of magnitude at least cd, and its second derivative is between positive constant multiples of d. Since $np^2 + p = o(d)$, the exact minimizing saddle stays in a fixed compact subinterval of $(0,1)$. Convexity makes it the global minimum over $s \ge 0$. In particular,

$$I^{\mathrm{cnt}}_{i,ab} := -\min_{s\ge 0} \Psi_{i,ab}(s) = J_i + O(np^2+p) = J_i + O(1). \tag{109}$$

One can also obtain this bound directly by taking minima in the uniform transform approximation.

Under the exact optimizing tilt, the Bernoulli probabilities on the two relevant groups are comparable to p. Their sum is $O(d)$, and the variance of $V_{i,ab}$ is comparable to d. Consequently lemma 2.1 applies to (107) without any spectral event. For every fixed $x_0 < \infty$, uniformly over $|x| \le x_0$,

$$c_{x_0} d^{-1/2} e^{-J_i} \le \mathbb{P}_a(V_{i,ab} < x) \le \mathbb{P}_a(V_{i,ab} \le x) \le C_{x_0} d^{-1/2} e^{-J_i}. \tag{110}$$

The strict lower bound from that lemma uses an open interval of fixed length under the tilt. Thus lattice atoms and an arbitrary fixed tie ordering cause no loss of the lower bound.

Comparison with the known-target likelihood. Let

$$\mathcal{T}_c = \sum_{j \neq i} \theta_j B_{c,z_j} = pT_c^{(i)}.$$

By the exact group normalization,

$$\begin{aligned} \mathcal{T}_a &= m[p + (K-1)q] - \theta_i p, \\ \mathcal{T}_b &= m[p + (K-1)q] - \theta_i q. \end{aligned} \tag{111}$$

In particular $\mathcal{T}_a, \mathcal{T}_b \asymp d$, $\mathcal{T}_a - \mathcal{T}_b = O(p)$, and $\log(\mathcal{T}_a/\mathcal{T}_b) = O(n^{-1})$.

Fix a sufficiently large constant L_0 to be chosen below, and consider the single-target event

$$\mathcal{E}_i = \{D_i \leq L_0 d\}.$$

The exact known-target Bernoulli contrast is

$$L_{i,\text{known}}^{ab} = V_{i,ab} + \sum_{j \neq i} (1 - A_{ij}) \log \frac{1 - \theta_i \theta_j B_{a,z_j}}{1 - \theta_i \theta_j B_{b,z_j}}.$$

Expansion of these nonedge logarithms on $\mathcal{E}_i$ gives

$$\begin{aligned} L_{i,\text{known}}^{ab} &= V_{i,ab} - \theta_i(\mathcal{T}_a - \mathcal{T}_b) + O(D_i p + np^2) \\ &= V_{i,ab} + O(np^2 + p). \end{aligned} \tag{112}$$

The $O(D_i p)$ term restores the incident-edge terms in the linear exposure; the quadratic remainder is $O(np^2)$.

Comparison with the capped Bernoulli profile. For a candidate c, put $b_{cj} = \theta_j B_{c,z_j}$. The admissible interval in (6) is

$$0 \leq t \leq t_{\max,c} := \frac{1}{2 \max_{j \neq i} b_{cj}}.$$

Because $b_{cj} \asymp p$, its upper endpoint tends to infinity. For $t > 0$ the derivative of the exact raw-row likelihood is

$$\frac{D_i}{t} - \sum_{j \neq i} (1 - A_{ij}) \frac{b_{cj}}{1 - t b_{cj}}.$$

On $\mathcal{E}_i$, there are at least $n - 1 - L_0 d \geq n/2$ nonedges for all sufficiently large n. Their total intensity is at least cd, uniformly over candidates. Hence the derivative is negative for every admissible t above a fixed constant depending only on L_0 and the structural bounds. Every exact maximizing t is therefore bounded. The cap at $1/2$ is inactive on this bounded interval for all large n.

Uniformly over the resulting bounded t interval, the likelihood is

$$D_i \log t + \sum_{j \neq i} A_{ij} \log b_{cj} - t\mathcal{T}_c + O(np^2). \tag{113}$$

Indeed, restoring the edge terms in the linear exposure costs $O(D_i p)$, and the sum of quadratic nonedge remainders costs $O(np^2)$. Both are $O(np^2)$ on $\mathcal{E}_i$. The leading optimizer $D_i/\mathcal{T}_c$ is bounded as well, so the same error bound holds after maximization. Terms $D_i \log D_i - D_i$ and $\sum_{j \neq i} A_{ij} \log \theta_j$ cancel in a contrast. Thus the exact capped profile satisfies

$$\begin{aligned} L_{i,\text{prof}}^{ab} &= V_{i,ab} - D_i \log(\mathcal{T}_a/\mathcal{T}_b) + O(np^2) \\ &= V_{i,ab} + O(np^2 + p) \quad \text{on } \mathcal{E}_i. \end{aligned} \tag{114}$$

For $D_i = 0$, the raw row is identically zero; $t = 0$ maximizes every candidate likelihood with value zero. The displayed formulas then hold with the continuous convention $0 \log 0 = 0$.

Single-row exceptions, directional errors, and averaging. Every mean degree is at most $\lambda_+ d$ for a fixed λ_+. For any prescribed fixed H, a Bernoulli Chernoff bound permits a fixed choice of L_0 such that

$$\sup_i \mathbb{P}(\mathcal{E}_i^c) \le e^{-Hd}. \tag{115}$$

Also $J_i \le C_J d$ uniformly. Choose $H > C_J + 1$. It follows from $d \to \infty$ that

$$\sup_i \frac{\mathbb{P}(\mathcal{E}_i^c)}{d^{-1/2} e^{-J_i}} \le \sqrt{d}\, e^{-(H - C_J)d} \longrightarrow 0.$$

This calculation concerns one target at a time; it introduces no factor n and requires no logarithmic lower bound on d.

By (112) and (114), a fixed constant B_0 bounds the difference between $V_{i,ab}$ and either exact contrast on $\mathcal{E}_i$. Thus for either $L_i^{ab} = L_{i,\mathrm{known}}^{ab}$ or $L_i^{ab} = L_{i,\mathrm{prof}}^{ab}$,

$$\begin{aligned} \mathbb{P}_a(V_{i,ab} < -B_0) - \mathbb{P}_a(\mathcal{E}_i^c) &\le \mathbb{P}_a(L_i^{ab} < 0) \\ &\le \mathbb{P}_a(L_i^{ab} \le 0) \\ &\le \mathbb{P}_a(V_{i,ab} \le B_0) + \mathbb{P}_a(\mathcal{E}_i^c). \end{aligned}$$

Applying (110) and then (115) proves that both the strict and nonstrict directional errors of either oracle are comparable to $d^{-1/2} e^{-J_i}$. These are unconditional event inclusions; no independence from $\mathcal{E}_i$ is asserted or needed.

For the raw block-count vote, the pairwise contrast is $V_{i,ab}$ itself. If counts are normalized by the self-excluded group sizes, multiplying all candidate scores by mL changes the pairwise contrast by at most $CLD_i/n = O(p)$ on $\mathcal{E}_i$, since those sizes are m or $m - 1$. The same tail comparison therefore applies to either voting convention.

For each of these coherent families of candidate scores, a strict pairwise defeat of the true group implies an error, and an error implies a nonstrict defeat by some competitor. Taking the maximum of the strict pair probabilities for a lower bound and their nonstrict sum for an upper bound proves that each per-target error is comparable to $d^{-1/2} e^{-J_i}$, since K is fixed. Averaging these bounds over i proves (106). □

Remark 5.12 (Oracle alignment and the algorithmic range). The degree-dependent exponent in (106) is the balanced DC-SBM exponent appearing in the minimax analysis of Gao, Ma, Zhang and Zhou [3]. Their exponent-level minimax expression does not, by itself, give the sharp prefactor or a constant-ratio theorem for every parameter point. The proposition supplies the exact local-oracle comparison on the fixed-separation, bounded-theta class stated here.

For (104), the finite-n exposures in (14) are exactly equal. Hence, on the overlap with the algorithmic range of theorem 1.2, corollary 1.3 and proposition 5.11 give

$$\mathbb{E}\ell(\widehat{z}_D, z) \le C R_{\theta,*}, \qquad R_{\theta,*} \asymp R_{\mathrm{prof},*}.$$

The absence of a logarithmic lower bound in the oracle proposition does not remove that bound from the spectral algorithm theorem. Extending the algorithm requires its own sparse spectral and likelihood-transfer estimates.

6 Paying every exceptional event at the oracle scale

The preceding sections prove fitted-score comparisons on events whose deterministic premises are consequences of raw noise, degree and row concentration bounds. This section completes the unconditional theorems. It is not enough that those events have probability tending to one.

6.1 Events with an arbitrarily large fixed exponential rate

For any fixed $H > 0$, choose the raw norm threshold $L_H\sqrt{d}$ in (23). Since $\lambda_i \le \lambda_+ d$ for a fixed λ_+, the elementary Bernoulli Chernoff inequality also gives, for a sufficiently large fixed b_H,

$$\mathbb{P}\{\max_i D_i > b_H d\} \le ne^{-Hd}.$$

For example the exponent in the Poisson upper bound is $b_H \log(b_H/\lambda_+) - b_H + \lambda_+$, which grows without bound as b_H grows. Here this is a bound on a Bernoulli transform, not a change of sampling model. The conditional events in lemma 3.1 contribute at most $2Kne^{-Hd}$. The constants in the deterministic lemmas may increase with H, but remain independent of n.

For DC-SBM we additionally impose

$$\max_{a,b} |[Z^\top (A - \Omega)Z]_{ab}| \le L_B n. \tag{116}$$

Each centered block total is a sum of independent bounded upper-triangular edge variables, with coefficients at most two and variance at most Cnd. The completed diagonal contributes only $O(d)$. Scalar Bernstein therefore implies, for sufficiently large fixed L_B and all large n,

$$\mathbb{P}\{(116) \text{ fails}\} \le 2K^2 \exp\left\{-\frac{cL_B^2 n}{d + L_B}\right\}.$$

Under $d^2 \le C_1 n$, the exponent is at least $c' L_B^2 d/C_1$ once $d \ge L_B$. Thus L_B can be chosen to make this probability at most $2K^2 e^{-Hd}$ for any prescribed fixed H. Ordinary free-profile fitting does not need this event: it integrates the block-fluctuation exponential moment instead, as proved in section 4.

Let $\mathcal{G}$ be the intersection of these required events and the minimum-degree event $D_i \ge a_0 d$. All spectral, initialization and fitting premises in the earlier sections follow on $\mathcal{G}$. In particular all tested deleted graphs satisfy their own measurable training premises there. Apart from the minimum-degree event, the failure probability is at most

$$C_K n e^{-Hd}. \tag{117}$$

6.2 Ordinary SBM on the explicit parameter class

For $P_{ab} \in [p/2, p]$ and $p \le 1/4$, the log-odds coefficient of any pairwise oracle contrast satisfies

$$|w_c^{ab}| \le W := \log(7/3).$$

Every tilted Bernoulli probability between the two candidates is at most p, so the second derivative of the log affinity in (20) is at most dW^2. A twice differentiable convex function vanishing at zero and one with second derivative at most M is at least $-Mt(1-t)/2$. Consequently

$$I_{i,ab} \le d\,\frac{\log^2(7/3)}{8} < 0.09d. \tag{118}$$

For all sufficiently large n, every mean degree is at least $0.49d$. Choose $a_0 = 1/16$. The Bernoulli lower-tail transform yields

$$\begin{aligned}\mathbb{P}(D_i < d/16) &\le \exp\left\{-d\left[0.49 - \frac{1}{16} - \frac{1}{16}\log(16 \cdot 0.49)\right]\right\} \\ &\le e^{-0.298d}.\end{aligned}$$

$$\mathbb{P}(\min_i D_i < d/16) \le ne^{-0.298d}. \tag{119}$$

Take $H = 1$, increasing the other fixed thresholds if necessary. Equations (117) and (119) give $\mathbb{P}(\mathcal{G}^c) \le Cne^{-0.298d}$. The information bound (118) applies to every vertex and every competitor. Thus the sharp pairwise lower bound (21), followed by averaging over all vertices, gives

$$R_{\mathrm{SBM},*} \ge \frac{1}{n}\sum_{i=1}^{n} \frac{c}{\sqrt{d}} e^{-0.09d} = \frac{c}{\sqrt{d}} e^{-0.09d}.$$

No additional balancing assumption is needed: the uniform coefficient bound, group-proportion bound and full-rank separation already hold at every vertex. Consequently

$$\frac{\mathbb{P}(\mathcal{G}^c)}{R_{\mathrm{SBM},*}} \le Cn\sqrt{d}\, e^{-0.208d}.$$

The function $\sqrt{d}\, e^{-0.208d}$ is decreasing once $d > 1/(2 \cdot 0.208)$. Therefore, for $d \ge 5 \log n$ and all sufficiently large n,

$$n\sqrt{d}\, e^{-0.208d} \le \sqrt{5 \log n}\, n^{1-5\cdot 0.208} = \sqrt{5 \log n}\, n^{-0.04} = o(1).$$

The earlier coefficient 20 was a sufficient numerical choice obtained by retaining only one vertex in the oracle lower bound and then bounding $\sqrt{d}$ by $\sqrt{n}$. Both losses are unnecessary. The improved coefficient 5 changes neither the encoder nor the likelihood decoder; it does not remove the global minimum-degree event from this proof. The good-event comparison in section 4 and (22) complete theorem 1.1.

6.3 The automatic margin for a general DC profile oracle

Let $J_{i,a\to b}$ be the exact Bernoulli directional profile information defined in section 5. For the self-excluded neighbor masses $M_c = \sum_{j\neq i : z_j = c} \theta_j$, set

$$T_a = \sum_c M_c C_{ac}, \qquad q_{ac} = \frac{M_c C_{ac}}{T_a}, \qquad h^*_{ab} = \min_{0\le s\le 1} \sum_c q_{ac}^{1-s} q_{bc}^{s}.$$

Uniform positive group proportions and entries of C give $q_{ac} \ge q_- > 0$ for all large n. Hence

$$h^*_{ab} \ge \sum_c \min(q_{ac}, q_{bc}) \ge K q_- > 0. \tag{120}$$

The exact log-transform expansion, uniformly in $s \in [0,1]$, is

$$\sum_{j\neq i} \log\{1 + \theta_i\theta_j p C_{az_j}(e^{-sw_{z_j}} - 1)\} = \theta_i p T_a \left[\sum_c q_{ac}^{1-s} q_{bc}^{s} - 1\right] + O(np^2).$$

Minimizing therefore gives

$$J_{i,a\to b} = \lambda_i(1 - h^*_{ab}) + O(np^2), \qquad \lambda_i = \theta_i p T_a. \tag{121}$$

This calculation uses the exact Bernoulli transform. Under $np^2 \le C_1$, (120) and $\lambda_i \ge \lambda_- d$ imply, for a fixed $\eta > 0$ and all large n,

$$J_{\min} := \min_i \min_{b\neq z_i} J_{i,z_i\to b} \le \lambda_{\min} - \eta d. \tag{122}$$

To see it, choose a vertex with minimum mean degree and any competitor. Equal candidate exposures are not required for this profile margin.

Choose $a_0 > 0$ sufficiently small that

$$a_0\{1 + \log(\lambda_+/a_0)\} \le \eta/2, \qquad a_0 < \lambda_-.$$

For each vertex, the Bernoulli lower-tail bound is

$$\mathbb{P}(D_i < a_0 d) \le \exp\{-\lambda_i + a_0 d + a_0 d \log(\lambda_i/(a_0 d))\}.$$

The exponent is monotone in λ_i above $a_0 d$. Together with (122), it follows that

$$\mathbb{P}(\min_i D_i < a_0 d) \le n \exp\{-J_{\min} - \eta d/2\}. \tag{123}$$

The full profile reference has the lower bound

$$R_{\mathrm{prof},*} \ge \frac{c}{n\sqrt{d}} e^{-J_{\min}}.$$

Thus the ratio of (123) to that reference is at most

$$C n^2 \sqrt{d}\, e^{-\eta d/2} = o(1)$$

when $d \ge C_0 \log n$ with, for example, $C_0 > 6/\eta$ and sufficiently large n. We used only $d \le n$ in bounding the polynomial factor. The constants can be enlarged without changing their independence of n.

Also $J_{\min} \le C_J d$ for a fixed C_J by its bounded-weight transform. Choosing $H > C_J + 1$ in (117), and enlarging C_0 if needed, pays those remaining failures at the same reference scale. Hence $\mathbb{P}(\mathcal{G}^c) = O(R_{\mathrm{prof},*})$. The good-event theorem and invariant-basin result in section 5 now prove theorem 1.2, including its finite hard-refit version.

Under the equal-exposure condition, the profile and known-target-degree oracle risks are comparable by the bounded-contrast proof in section 5. This proves the DC assertion of corollary 1.3. For the native Euclidean ordinary-SBM decoder, the same sparse transform expansion gives $I_{i,ab} = \lambda_i(1 - h^*_{ab}) + O(np^2 + p)$ under equal exposures. Thus its information–degree margin is also automatic. The general-positive-profile version of section 4, with its block fluctuation integrated instead of imposing (116), therefore proves the remaining assertion of corollary 1.3.

6.4 An explicit extension for ordinary SBM with unequal exposures

The entry-ratio-two restriction in theorem 1.1 is a simple sufficient way to ensure the failure-event comparison. The deterministic and conditional likelihood proof covers a larger parameter range.

Proposition 6.1 (A deterministic information–degree margin suffices)**.** *Suppose ordinary SBM has fixed K, positive comparable group sizes, $c_- p \le P_{ab} \le c_+ p$, $\max P_{ab} \le 1 - \epsilon_0$, and $\sigma_{\min}(P) \ge \kappa p$, for fixed positive constants. If for some fixed $\eta > 0$ and all sufficiently large n,*

$$\lambda_{\min} - I_{\min} \ge \eta d, \qquad d = np \ge C_0 \log n,$$

then the native Euclidean signed likelihood decoder, with sufficiently small structural floor coefficient, satisfies $\mathbb{E}\ell(\widehat{z}_E, z) \le C R_{\mathrm{SBM},*}$ when C_0 is sufficiently large. No condition $np^2 = O(1)$ is needed for this ordinary-SBM proposition.

Proof. All true Bernoulli likelihood coefficients and tilted means are exactly block constant, so the Euclidean projection, fitted-score and sharp-tail arguments of section 4 apply without a sparse Bernoulli approximation. Generated profiles remain comparable to p and uniformly away from one when the floor is chosen for the structural class. The displayed margin, the same small degree cutoff used above and an arbitrarily strong choice of the other exponential bounds imply $\mathbb{P}(\mathcal{G}^c) = O(R_{\mathrm{SBM},*})$. Here the polynomial comparison uses $d \le Cn$, which follows from $c_- p \le P_{ab} \le 1 - \epsilon_0$; it does not require $p \le 1$. □

This proposition is a condition on the deterministic model parameters, not an assumed success event or a hypothesis that the desired risk ratio already holds. It is nevertheless an additional restriction. For model families violating it, a refined analysis of exceptional low-degree graphs or a different normalized encoder is needed; the crude loss bound one on all such graphs does not prove a constant ratio.

7 Why the known-target oracle cannot be the general DC-SBM benchmark

The profile reference and an oracle supplied with the target's true degree parameter are different experiments. The distinction cannot be removed by a more accurate spectral reconstruction: the following obstruction holds even when the classifier observes the entire adjacency matrix. The example has bounded positive degree parameters, exact block normalization, balanced groups, a positive definite block matrix, and logarithmic expected degree. Every probability calculation below is for the Bernoulli graph itself.

The local known-target reference. For a fixed DC-SBM parameter point $\tau = (z^\tau, \theta^\tau)$, write $\mathbb{P}_\tau$ for the graph law. At a target vertex v, give a local oracle the values of p, C, θ_v^τ and all neighbor parameters $(z_j^\tau, \theta_j^\tau)$, $j \neq v$. Its candidate row laws are

$$q_{vj}^{(a,\tau)} = p\theta_v^\tau \theta_j^\tau C_{az_j^\tau}, \qquad L_v^{(a,\tau)}(A_{v,-v}) = \prod_{j \neq v} (q_{vj}^{(a,\tau)})^{A_{vj}} (1 - q_{vj}^{(a,\tau)})^{1-A_{vj}}. \tag{124}$$

Let $\delta_v^{\mathrm{known},\tau}$ maximize these two likelihoods with equal candidate priors and a fixed tie rule, and define

$$R_{\mathrm{known}}^\tau(n) = \frac{1}{n} \sum_{v=1}^n \mathbb{P}_\tau\{\delta_v^{\mathrm{known},\tau} \neq z_v^\tau\}. \tag{125}$$

The target parameter is held at its supplied true value under both candidate labels. The local comparison does not impose a global exact-group-size or block-normalization constraint after revealing the other labels; an exact group-size constraint would itself determine a missing label. Thus (125) is the nondegenerate local likelihood benchmark, not a globally constrained oracle experiment.

Theorem 7.1 (An exact Bernoulli obstruction for the known-target oracle)**.** *Let $n = 2m$ tend to infinity through even integers, and set*

$$C = \begin{pmatrix} 4 & 2 \\ 2 & 11/10 \end{pmatrix}, \qquad p = \frac{M \log n}{n}, \qquad M = 100. \tag{126}$$

There are two explicit parameter sequences, denoted $+$ and $-$, for the independent-edge Bernoulli DC-SBM with probabilities $p\theta_u\theta_v C_{z_u z_v}$, such that both sequences satisfy

$$n_1 = n_2 = m, \qquad \sum_{v: z_v = a} \theta_v = m \quad (a = 1, 2), \qquad \frac{1}{2} \leq \theta_v \leq \frac{3}{2}. \tag{127}$$

For every possibly randomized clustering algorithm based on the full graph and information common to the two parameter points, put

$$R_{\mathrm{alg}}^\tau(n) = \mathbb{E}_\tau \left[\frac{1}{n} \min_{\sigma \in \mathfrak{S}_2} \sum_{v=1}^n \mathbf{1}\{\sigma(\widehat{z}_v) \neq z_v^\tau\} \right], \qquad \tau \in \{+, -\}. \tag{128}$$

Define

$$c = \frac{(2 - \sqrt{2})^2}{8}, \qquad \gamma = \frac{(\sqrt{11/10} - 1)^2}{2}. \tag{129}$$

Then

$$\max_{\tau \in \{+,-\}} \frac{R_{\mathrm{alg}}^\tau(n)}{R_{\mathrm{known}}^\tau(n)} \geq \frac{1 + o(1)}{4} \, n^{M(c-2\gamma)-1}. \tag{130}$$

The exponent is strictly positive; at $M = 100$,

$$M(c - 2\gamma) - 1 = 3.05109151537556\ldots > 0. \tag{131}$$

In particular, no algorithm with unknown vertex degree parameters can have a bounded risk ratio to (125) *along every admissible parameter sequence in this class. This includes every algorithm based only on retained spectral information and observed degrees.*

Proof. **Construction and exact normalization.** Distinguish vertices $i = 1$ and $k = 2$. Partition the remaining vertices into two fixed anchor sets V_1, V_2, each of size $m - 1$. Every anchor in V_a has label a under both parameter points. Give anchors in V_1 the common degree parameter

$$t_m = 1 + \frac{1}{2(m-1)},$$

and anchors in V_2 degree parameter one. At the two parameter points set

$$\begin{aligned} +: &\quad (z_i, \theta_i) = (1, 1/2), \quad (z_k, \theta_k) = (2, 1), \\ -: &\quad (z_i, \theta_i) = (2, 1), \quad\quad (z_k, \theta_k) = (1, 1/2). \end{aligned} \tag{132}$$

Each group has exactly m vertices. Its total degree parameter is exactly m, since

$$(m-1)t_m + \frac{1}{2} = m, \qquad (m-1) + 1 = m.$$

The stated degree bounds hold for $m \geq 2$. The matrix C has positive entries, positive trace, and determinant $2/5 > 0$, so it is positive definite with a fixed positive smallest eigenvalue. All edge probabilities belong to $(0, 1)$ for sufficiently large n.

Only $n - 2$ Bernoulli coordinates distinguish the graph laws. All anchor–anchor probabilities agree between the two parameter points. The edge i–k has probability p under either point. For an edge from either distinguished vertex to an anchor of V_1, the two candidate intensities agree exactly:

$$\frac{1}{2} \cdot 4 \cdot t_m = 1 \cdot 2 \cdot t_m = 2t_m.$$

Thus the only distinct coordinates are the $2(m-1) = n - 2$ edges from $\{i, k\}$ to V_2. Their two probabilities are p and $(11/10)p$, exchanged between $+$ and $-$.

For probability mass functions P, Q, let $\mathcal{A}(P, Q) = \sum_x \sqrt{P(x)Q(x)}$ be their affinity. Independence gives the exact graph affinity

$$\mathcal{A}_n = \mathcal{A}(\mathbb{P}_+, \mathbb{P}_-) = \left[\sqrt{11/10}\, p + \sqrt{(1-p)(1-(11/10)p)}\right]^{n-2}. \tag{133}$$

The factor in brackets has the elementary expansion

$$\sqrt{11/10}\, p + \sqrt{(1-p)(1-(11/10)p)} = 1 - \gamma p + O(p^2).$$

Since $np^2 = M^2(\log n)^2/n = o(1)$,

$$\mathcal{A}_n^2 = \exp\{-2M\gamma \log n + o(1)\} = (1 + o(1))n^{-2M\gamma}. \tag{134}$$

The elementary Cauchy–Schwarz bound $\mathrm{TV}(P, Q) \leq \sqrt{1 - \mathcal{A}(P, Q)^2}$ implies that every test ϕ of the two graph laws satisfies

$$\frac{\mathbb{P}_+(\phi = -) + \mathbb{P}_-(\phi = +)}{2} \geq \frac{1 - \mathrm{TV}(\mathbb{P}_+, \mathbb{P}_-)}{2} \tag{135}$$

$$\geq \frac{\mathcal{A}_n^2}{4} = \frac{1 + o(1)}{4} n^{-2M\gamma}, \tag{136}$$

where the second inequality uses $1 - \sqrt{1 - y} \geq y/2$. Independent randomization by the test or the clustering algorithm does not alter this bound: adjoining a common random seed leaves the affinity unchanged.

From testing to graph-average permutation loss. Orient the two output labels of an arbitrary estimated partition by the permutation minimizing disagreement with the fixed anchor labels, using a fixed tie rule. This is one measurable rule under both parameter points, since the anchors and their labels are the same. Define a test ϕ to declare $+$ if the oriented label of vertex i is 1, and $-$ otherwise.

If the estimated partition has zero permutation-invariant error, the anchor orientation recovers its correct permutation uniquely: both anchor sets are nonempty. Therefore an erroneous test outcome implies at least one error in the permutation-invariant count. Pathwise, under either point,

$$\frac{1}{n}\min_{\sigma\in\mathfrak{S}_2}\sum_v \mathbf{1}\{\sigma(\widehat{z}_v)\neq z_v^\tau\}\geq \frac{1}{n}\mathbf{1}\{\phi\neq\tau\}. \tag{137}$$

Taking expectations and using (136) gives

$$\frac{R^+_{\mathrm{alg}}(n)+R^-_{\mathrm{alg}}(n)}{2}\geq \frac{1+o(1)}{4}n^{-2M\gamma-1}. \tag{138}$$

The anchors are needed only to turn a partition into a test in this proof. Indeed, their labels and degree parameters, the block matrix, the exact group sizes, and the block-normalization constraints could all be revealed to the clustering algorithm: they are common to the two parameter points, and (136) would still apply.

An upper bound on every known-target local error. For any target v, at least $m-2$ anchors in V_1 are different from v. For each such anchor j, its two local candidate edge probabilities in (124) are

$$u=4p\theta_v\theta_j,\qquad v'=2p\theta_v\theta_j.$$

They satisfy

$$(\sqrt{u}-\sqrt{v'})^2=p\theta_v\theta_j(2-\sqrt{2})^2\geq\frac{p}{2}(2-\sqrt{2})^2.$$

For any Bernoulli probabilities u,v', the arithmetic–geometric mean inequality for their nonedge probabilities yields

$$\sqrt{uv'}+\sqrt{(1-u)(1-v')}\leq 1-\frac{(\sqrt{u}-\sqrt{v'})^2}{2}\leq\exp\left\{-\frac{(\sqrt{u}-\sqrt{v'})^2}{2}\right\}. \tag{139}$$

Under either local true label, Markov's inequality applied to the square root of the competing likelihood ratio bounds the LRT's directional error by the product affinity of all its row coordinates. Discarding coordinates outside these $m-2$ anchors can only increase that upper bound. Hence, uniformly over targets and both parameter points,

$$\mathbb{P}_\tau\{\delta_v^{\mathrm{known},\tau}\neq z_v^\tau\}\leq\exp\left\{-\frac{(m-2)p}{4}(2-\sqrt{2})^2\right\} \tag{140}$$

$$=(1+o(1))n^{-Mc}. \tag{141}$$

In particular,

$$R^+_{\mathrm{known}}(n)+R^-_{\mathrm{known}}(n)\leq 2(1+o(1))n^{-Mc}. \tag{142}$$

Finally, the maximum of two nonnegative risk ratios is at least their denominator-weighted average. Combining (138) and (142) therefore proves (130). The constants in (129) are

$$c=0.04289321881345246\ldots,\qquad \gamma=0.00119115182984846\ldots,$$

which give the exponent in (131). Its positivity does not depend on numerical evaluation: $c>1/25$ and $2\gamma<1/400$, so $100(c-2\gamma)-1>11/4>0$. □

Remark 7.2 (Scope of the obstruction). The same proof works for every fixed $M > (c-2\gamma)^{-1}$, so its logarithmic-degree multiplier can exceed any prescribed fixed lower threshold. The example also satisfies $np^2 \to 0$. Thus neither a larger logarithmic expected degree nor the sparse Bernoulli approximation removes the nuisance-information gap.

The result concerns a uniform comparison to the known-target reference, not the full-edge profile benchmark used by the positive DC-SBM theorem. It also does not assert that every fixed DC-SBM has an unbounded ratio: it proves that at least one of the two explicit parameter sequences has an unbounded limsup ratio for any algorithm. Additional restrictions can make the stronger benchmark attainable. For example, the balanced block strengths in this counterexample are $C(1/2, 1/2)^T = (3, 31/20)^T$, which are unequal; the example lies outside the equal-exposure corollary.

The counterexample is also outside the assortative parameter class of Gao, Ma, Zhang and Zhou [3]: its smallest diagonal entry is 1.1, whereas its off-diagonal entry is 2. Their class requires every within-group strength to exceed every between-group strength. Therefore this obstruction is not a negative result for their classical balanced planted-partition benchmark. That benchmark is explicitly covered by proposition 5.11 on the stated bounded-parameter class. Their classwise minimax theorem and a pointwise comparison to a known-target oracle on every arbitrary block matrix are different claims.

Finally, a lower bound for an experiment retaining only block counts would not by itself establish a lower bound for arbitrary data-dependent real spectral coordinates. The present theorem needs no such compression claim: it is a lower bound for the full adjacency experiment with unknown target degree parameters.

8 Scope, implementation changes, and proof audit

8.1 Results established in this note

Model and decoder	Reference risk	Proved model range
Ordinary SBM; Euclidean signed free-profile likelihood	Full-edge true-parameter Bernoulli row LRT	Theorem 1.1; more generally proposition 6.1, including sparse equal exposures with arbitrary fixed positive full-rank profiles.
DC-SBM; canonical degree-metric reconstruction and fitted profile LRT	Full-edge profile LRT with true neighbor parameters	General positive full-rank bounded-parameter class, fixed K, $C_0 \log n \le np$ and $np^2 \le C_1$.
DC-SBM; the same degree-metric procedure	Full-edge LRT additionally supplied with the target's true degree	The preceding sparse class plus finite-n equal block exposures; the classical balanced planted-partition local oracle is included.
General DC-SBM; any graph algorithm	Known-target-degree local LRT	A uniform constant ratio is impossible; theorem 7.1 supplies two explicit normalized parameter sequences.

The algorithmic positive statements are bounds on expected misclassification, with failure events included. None is merely a statement conditional on correct initial groups, small fitted-parameter error, a favorable LOO projector, or successful numerical k-means.

8.2 What changed relative to the recent experimental code

The recent ordinary implementation evaluates the full Bernoulli working criterion on signed $A\Pi_X$, uses finite growing/replacement trials, and performs the mandatory final E/M step. The ordinary theorem analyzes that mathematical schedule. For arbitrary fixed K the encoder is

parameterized by K; a script hardcoded to three eigenvectors is just its $K = 3$ case. If the signal matrix is indefinite, largest absolute eigenvalues are required. Selecting largest algebraic eigenvalues is covered only when all informative population eigenvalues are positive.

For DC-SBM, $A\Pi_X$ and AH_D are different reconstructions. The positive theorem here uses the degree-metric action AH_D and its explicit hollow correction. It must not be retrospectively attached to an experimental curve computed with the prior Euclidean action. The retained input (D, U, Λ) is identical; the decoder changes. The radius-count initializer is a concrete certified alternative to unquantified fixed-restart Lloyd clustering. A certified constant-factor k-means initializer could also be used, but its own failure probability would have to be paid at the oracle scale.

The theorems use exact algebraic spaces and exact arithmetic, including the specified exact E/M updates for the ordinary decoder and the stated block fits and target-profile maxima for DC-SBM. The ordinary finite schedule does not assume a global maximizer of the mixture likelihood. They do not assert that a fixed floating-point tolerance remains adequate as $n \to \infty$. In particular, augmenting $D^{-1/2}U$ by an unscaled constant column and applying a fixed relative rank cutoff is not an asymptotic definition of its algebraic rank. Because the constant vector is already retained, it can be omitted from that numerical augmentation, or columns can be scaled before orthogonalization. Quantified numerical errors may be admitted if their induced likelihood-contrast and cumulant errors are uniformly bounded.

8.3 Why this is a risk bound

The error incurred in replacing a full-graph spectral score by its LOO version is deterministically $O(1)$ on the proved spectral event. Fitted nuisance parameters cost $O(1)$ in the DC contrast; ordinary free profiles produce $O(1 + \mathcal{B}_i)$ with a bounded exponential moment. The remaining projection error against the raw oracle row is random: it is not asserted to be uniformly bounded for every possible row. Instead its effective coefficients are bounded, its mean under the oracle error-boundary tilt is bounded, and its tilted quadratic moment is bounded. Lemma 2.1 converts precisely those facts into a constant risk multiplier while preserving $d^{-1/2}e^{-I}$.

The weighted cancellation controls the first-order mean term that a generic unweighted dot product need not control in DC-SBM. It does not assert exact recovery of every individual edge or of every observed block count. The Bernoulli profile reduction has a visible $O(np^2)$ remainder; extending the DC theorem beyond bounded np^2 requires a new argument for that remainder and the heterogeneous nonedge weights.

The more sparse algorithmic range $np = o(\log n)$, a vanishing normalized signal singular value, or an unrestricted known-θ_i oracle in general DC-SBM is not covered by the positive statements. The first two are unproved extensions here; the last is explicitly impossible by theorem 7.1. None should be folded into the present theorem by an unstated assumption or a change of denominator. The oracle comparison in proposition 5.11 does already hold below logarithmic degree. Section 9 separately proves the regularized geometry and a more flexible sharp transfer lemma, while identifying the outstanding sparse algorithmic conditions.

8.4 Audit record

The generalized-eigenvector full-to-LOO derivation was checked independently against a second argument based on a normalized arrowhead decomposition and a residual estimate for the degree-renormalized minor. The row-incoherence bootstrap uses a verified weighted Bernoulli concentration statement and controls every deleted basis simultaneously. The finite ordinary likelihood schedule was checked against the current implementation, including signed observations, trial resets, strict replacement acceptance and the mandatory endpoint order. The DC proof was separately checked for the uniform effect of data-dependent label errors, exact Bernoulli block fitting, the target profile cap, degree-zero oracle rows and adaptive finite hard refits.

The two-point obstruction was verified independently from its affinity and permutation-loss calculations.

These are internal mathematical and implementation checks, not external peer review. The note provides the complete proofs so each claim can be reviewed directly. There is no simulation estimate being used as proof of a bounded asymptotic ratio.

9 Sparse extension: established ingredients and the remaining theorem

The coefficient 20 was not a recovery threshold. The improved probability calculation in section 6 gives 5 for the same ordinary decoder. Removing the entire logarithmic lower bound is a different step: the existing encoder convention rejects a graph with any zero degree, and that convention cannot simply be retained below logarithmic expected degree. The sparse construction already present in the project avoids this algebraic difficulty. This section preserves that construction, proves additional degree-floor and sharp-transfer facts, and states precisely which algorithmic estimates remain to be proved. The results here do not silently extend theorems 1.1 and 1.2.

9.1 The existing degree-plus-spectrum action remains valid

Write $\mathbf{d} = A\mathbf{1}$ for the raw degree vector. For any strictly positive diagonal matrix W, retain

$$WAWU = U\Lambda, \qquad U^\top U = I, \qquad X = [\mathbf{1}, WU], \qquad Y = [\mathbf{d}, W^{-1}U\Lambda].$$

Then the exact identities

$$AX = Y, \qquad YX^\dagger = A\Pi_X, \qquad A\Pi_X\mathbf{1} = \mathbf{d} \tag{143}$$

hold for every graph, including graphs with zero degrees. Here Π_X is the Euclidean orthogonal projector onto the column space of X; no full-column-rank assumption is necessary. The earlier choice $W_{ii} = \min(1, 2\bar{D}/D_i)$ for $D_i > 0$, with $W_{ii} = 1$ for $D_i = 0$ and $\bar{D} = n^{-1}\sum_i D_i$, supplies a positive reweighting when $\bar{D} > 0$. The all-zero graph can be assigned a fixed output. The block-count error is exactly $-A(I - \Pi_X)Z$.

These facts need no logarithmic expected-degree assumption. They are computational identities, however: their validity does not give the sharp tail of the remaining random block-count error. In DC-SBM a normalization must also preserve the effective likelihood coefficients, as opposed to preserving only a population mean.

9.2 Degree flooring preserves the relevant DC geometry

Use the completed mean from (2) and put $D_0 = \mathrm{diag}(\Omega\mathbf{1})$. Its diagonal is between fixed positive multiples of d. Choose a fixed $a_0 > 0$ below its minimum structural multiplier and define

$$G_i = \max(D_i, a_0 d), \qquad G = \mathrm{diag}(G_i), \qquad S_G = G^{-1/2}AG^{-1/2}.$$

At the deterministic population degrees the floor is inactive. In particular $D_0^{-1}\Theta Z$ is block constant; adding a multiple of d to every degree would not have this property for heterogeneous theta.

Proposition 9.1 (Sparse degree-floor geometry)**.** *Under the fixed structural conditions (2)–(3), suppose $d = np \to \infty$ and $np^2 \le C_1$. For every fixed $r > 0$, with probability at least $1 - C_r n^{-r}$,*

$$\left\|G^{-1/2}(A - \Omega)G^{-1/2}\right\| \le C_r d^{-1/2}, \qquad \sum_i G_i \le C_r nd, \tag{144}$$

$$\sum_i \frac{(D_i - (D_0)_{ii})^2}{G_i} + \sum_i \frac{(G_i - (D_0)_{ii})^2}{G_i} \le C_r n. \tag{145}$$

Let U contain the K eigenvectors of S_G with largest absolute eigenvalues, let $\Pi = UU^\top$, and let $H = G^{-1/2}\Pi G^{1/2}$. For all sufficiently large n, on the same event the retained eigenvalues have a fixed separation from zero and

$$\left\|(I-\Pi)G^{1/2}Z\right\|_F \le C_r\sqrt{n}. \tag{146}$$

For every block-constant f with $\|f\|_\infty \le C_f$,

$$\|(I-H)f\|_G \le C_r\sqrt{n}, \qquad \|(I-H)f\|_2^2 \le C_r n/d, \quad \|v\|_G^2 = v^\top G v. \tag{147}$$

No lower bound proportional to $\log n$ is imposed.

Proof. Let $d_* = \max_{i\neq j} n\,\mathbb{E}A_{ij} \asymp d$, and set $W_{h,ii} = \sqrt{\min(1, 2d_*/D_i)}$, with value one at zero degrees. The changed set $\mathcal{S} = \{i : D_i > 2d_*\}$ has at most $10n/d_*$ vertices except on an event of probability e^{-cn} for large n. Here is a direct verification of the probability strength. If $m = \lfloor 10n/d_* \rfloor + 1 \le n$ vertices all exceed that degree threshold, some set T of size m has $\sum_{i\in T} D_i > 2md_*$. This sum is an independent-edge sum with coefficients in $\{1,2\}$ and mean at most md_*. At $t = (\log 2)/2$ its Bernoulli exponential transform gives a tail at most

$$\exp\{-(\log 2 - 1/2)md_*\}.$$

Indeed $(e^{tc}-1)/c \le (e^{2t}-1)/2 = 1/2$ for $c = 1, 2$. The logarithm of the number of sets is $\log\binom{n}{m} \le m\log(en/m) = o(md_*)$, because $d_* \to \infty$. This proves the asserted e^{-cn} bound. If $m > n$ there is no exceptional set to consider.

The squared Euclidean norm of every row of W_hAW_h is at most $W_{h,ii}^2 D_i \le 2d_*$. Apply the row-norm form of the regularized adjacency theorem of Le, Levina and Vershynin [4, Theorem 2.1 and Remark 2.2]. For a fixed sufficiently large choice of its probability parameter, it gives

$$\|W_hAW_h - \mathbb{E}A\| \le C_r\sqrt{d}$$

with failure probability at most $C_r n^{-r}$. The weights are allowed to depend on the graph. Also

$$\|W_h\Omega W_h - \Omega\| \le Cp\sqrt{n|\mathcal{S}|} \le C\sqrt{d};$$

the matrix on the left is supported on rows or columns in $\mathcal{S}$, so this follows already from its Frobenius norm. The completed-mean diagonal has norm $O(p)$. These observations imply $\|W_h(A-\Omega)W_h\| \le C_r\sqrt{d}$.

There is a diagonal F with bounded norm, depending only on the structural constants and a_0, such that

$$G^{-1/2} = (2d_*)^{-1/2}FW_h.$$

For $D_i > 2d_*$ its diagonal entry is one, and otherwise it is $\sqrt{2d_*/G_i}$. Conjugating the preceding bound proves the first part of (144). Independently, the Bernoulli transform of the total number of edges shows $\sum_i D_i \le Cnd$ except with probability e^{-cnd}. Since $G_i \le D_i + a_0 d$, this proves the second part.

Apply the normalized noise to $G^{1/2}\mathbf{1}$:

$$\left\|G^{-1/2}(A-\Omega)\mathbf{1}\right\|_2 \le C_r d^{-1/2}\left\|G^{1/2}\mathbf{1}\right\|_2 \le C_r\sqrt{n}.$$

This proves the first sum in (145). Because $a_0 d < (D_0)_{ii}$, flooring can only reduce the distance to $(D_0)_{ii}$: $|G_i - (D_0)_{ii}| \le |D_i - (D_0)_{ii}|$. This proves the second sum without any separate degree-wise union bound.

The nonzero eigenvalues of $G^{-1/2}\Omega G^{-1/2}$ have a fixed gap from zero. To verify this without a maximum-degree assumption, its factor $V = G^{-1/2}\Theta Z$ has diagonal Gram entries

$$\frac{cn}{d} \le \frac{(\sum_{i:z_i=a}\theta_i)^2}{\sum_{i:z_i=a}G_i} \le \sum_{i:z_i=a}\frac{\theta_i^2}{G_i} \le \frac{Cn}{d}.$$

The lower bound uses comparable group sizes and the total-degree upper bound; the upper bound uses the floor. Thus the nonzero spectrum is that of $p(V^\top V)^{1/2}C(V^\top V)^{1/2}$ and is separated from zero by a fixed amount. Spectral perturbation and (144) give projector error $O(d^{-1/2})$.

Now $G^{-1/2}D_0Z$ belongs to the population signal space, whereas

$$\left\|G^{1/2}Z - G^{-1/2}D_0Z\right\|_F^2 = \sum_i \frac{(G_i - (D_0)_{ii})^2}{G_i} \le C_r n.$$

Moreover $\left\|G^{1/2}Z\right\|_F^2 = \sum_i G_i \le C_r nd$. Combining these facts with the projector error proves (146). Since H is the G-orthogonal projector corresponding to Π, the first bound in (147) follows. The second uses $G_i \ge a_0 d$. □

The scale d in this proof need not be an algorithm input. One may use a sufficiently small fixed multiple of $\bar{D}$ for the floor. The same total-edge Bernoulli transform also gives the lower bound $\sum_i D_i \ge cnd$ with failure probability $e^{-c'nd}$. On this two-sided total-edge event, the threshold lies between fixed positive multiples of d and below every $(D_0)_{ii}$. Every deterministic step above then applies. The formula is defined even at isolated vertices. Deleting a vertex and recomputing degrees gives the same statements on the remaining graph, with its completed mean and block masses. Choosing the probability parameter larger also permits a union over all such deleted graphs. This does not yet compare their scores to the full graph score.

9.3 The DC tilted mean and quadratic error already close

Work on a deleted graph so that all coordinates below index actual neighbors of a separate tested vertex. For the bounded block-constant linear candidate contrast used in the DC profile reduction, the exact tilted Bernoulli probabilities have the decomposition

$$r = D_0 h + \epsilon, \qquad h \text{ block constant}, \quad \|h\|_\infty \le C/n, \quad \|\epsilon\|_\infty \le Cp^2. \tag{148}$$

For example the leading term of the exact tilt is $\theta_i\theta_j p C_{a,z_j} e^{-sw_{z_j}}$, while $(D_0)_{jj} = \theta_j p \sum_k \theta_k C_{z_j,z_k}$. Dividing the former by the latter gives the required block-constant h_j. The denominator in the exact Bernoulli tilt contributes $O(p^2)$ uniformly. This uses the original Bernoulli law throughout. The target coordinate is absent; artificially appending $r_i = 0$ would instead create an $O(p)$ error at that one coordinate.

For bounded block-constant f, put $e = (H - I)f$. The G-orthogonality of H gives the exact cancellation

$$r^\top e = -\langle (I-H)h, (I-H)f\rangle_G + ((D_0 - G)h)^\top e + \epsilon^\top e. \tag{149}$$

The first term is $O(1)$ by (146), since the block coefficients of h are $O(1/n)$. For the second, Cauchy–Schwarz and (145) give

$$|((D_0 - G)h)^\top e| \le \|h\|_\infty \left(\sum_j \frac{((D_0)_{jj} - G_j)^2}{G_j}\right)^{1/2} \|e\|_G \le C.$$

The last term is at most $Cp^2\sqrt{n}\,\sqrt{n/d} \le Cnp^2$. Also $r_j \le Cp$, so

$$|r^\top e| \le C(1 + np^2), \qquad \sum_j r_j e_j^2 \le Cp\,\|e\|_2^2 \le C. \tag{150}$$

Thus the first-order bias cancellation and bounded quadratic error do not require logarithmic degree. A uniform coefficient bound, or a suitable replacement for it, is still needed for sharp risk transfer.

9.4 Exact preservation of the raw degree vector

Let $S_G U = U\Lambda$, put $H = G^{-1/2}UU^\top G^{1/2}$, and define

$$R_0 = G^{1/2}U\Lambda U^\top G^{1/2} = AH, \qquad R = R_0 + (\mathbf{d} - R_0\mathbf{1})\frac{\mathbf{1}^\top}{n}. \tag{151}$$

All quantities in this formula are computable from the retained degrees and eigenpairs. Exactly $R\mathbf{1} = \mathbf{d}$. For every bounded block-constant likelihood vector w, with $\bar{w} = n^{-1}\mathbf{1}^\top w$,

$$Rw = A\{H(w - \bar{w}\mathbf{1}) + \bar{w}\mathbf{1}\}. \tag{152}$$

Hence the coefficient error is precisely $(H - I)(w - \bar{w}\mathbf{1})$, to which (150) applies on a deleted graph. This construction avoids orthogonalizing a possibly tiny residual constant vector. We do not assert that R is an orthogonal-projection reconstruction.

A hollow version subtracts R_{ii} from entry (i, i) and adds $R_{ii}/(n-1)$ to each other entry of row i. This preserves raw degrees and changes a bounded row contrast by at most $2\|w\|_\infty |R_{ii}|$. A sufficiently strong local bound for R_{ii} must therefore be proved before this hollowing can be inserted into a sparse algorithmic theorem; it does not follow from aggregate geometry alone.

9.5 Two sharp transfers that do not assume every coefficient is bounded

The next two lemmas give concrete alternatives to a global row-incoherence event. They concern a row-independent, training-measurable score, just as in lemma 2.1.

Lemma 9.2 (Exponential-moment coefficient control)**.** *Assume the oracle hypotheses of lemma 2.1. Let $\widehat{T} = T + b + \sum_j e_j X_j$, with (b, e) measurable from independent training data. On a measurable training event suppose*

$$\left| b + \sum_j r_j e_j \right| \le B, \qquad \sum_j r_j e_j^2 e^{|e_j|} \le V,$$

where B, V are fixed. Then, for all real x and all sufficiently large d,

$$\mathbb{P}(\widehat{T} \le x \mid \textit{training}) \le Ce^{s_* x + B + V/2} d^{-1/2} e^{-I}.$$

The constant is independent of the largest coefficient.

Proof. Taylor's inequality gives, under the original oracle tilt Q,

$$\log \mathbb{E}e^{-s_*\widehat{T}} \le -I - s_*\left(b + \sum_j r_j e_j\right) + \frac{1}{2}\sum_j r_j e_j^2 e^{|e_j|} \le -I + B + V/2.$$

Tilt again by $e^{-s_*\widehat{T}}$. The new probabilities satisfy

$$\widehat{r}_j = \frac{r_j e^{-s_* e_j}}{1 - r_j + r_j e^{-s_* e_j}} \le r_j e^{s_*|e_j|}.$$

For $|e_j| \le 1$, this bounds the new centered third absolute moments by $C_W \sum_j r_j = O(d)$. For $|e_j| > 1$, use $s_* \le 1 - \tau$ and

$$e^{s_*|u|}(W + |u|)^3 \le C_{W,\tau} u^2 e^{|u|} \qquad (|u| > 1).$$

The remaining third moments are therefore $O(V)$. The upper variance bound follows from the same estimates. There is no restriction that every e_j be bounded.

For the lower variance, restrict to $|e_j| \le 1$. The original variance discarded by this restriction is at most W^2V, and the weighted ℓ_2 norm of the remaining perturbation is at most $\sqrt{V}$. Thus the square root of the original-weight variance of the remaining $(w_j + e_j)$ is at least $\sqrt{vd - W^2V} - \sqrt{V}$. On this restricted set, the new and old Bernoulli variances differ by only fixed factors. The new total variance is therefore at least cd for large d. Berry–Esseen gives a $C/\sqrt{d}$ bound for every unit interval under the new tilt. Summing those intervals, using the first displayed exponential-moment bound, proves the assertion as in lemma 2.1. □

Lemma 9.3 (Paying exceptional coefficients only when the row hits them)**.** *Assume the oracle hypotheses of lemma 2.1, and write p_j for the un-tilted row probabilities. Suppose $p_j \le C_p r_j$, as holds here for bounded oracle likelihood coefficients. On an independent training event $\mathcal{E}$, suppose*

$$\left| b + \sum_j r_j e_j \right| \le B, \qquad \sum_j r_j e_j^2 \le V.$$

Here B, V are fixed constants. For any fixed $M > 0$ define

$$h_M = \sum_{j:|e_j|>M} p_j, \qquad \eta_M = \sum_j r_j e_j^2 \mathbf{1}\{|e_j| > M\}.$$

Then, for every fixed bounded x,

$$\mathbb{P}(\widehat{T} \le x) \le C_M d^{-1/2} e^{-I} + \mathbb{E}[\mathbf{1}\{\mathcal{E}\} h_M] + \mathbb{P}(\mathcal{E}^c). \tag{153}$$

Since $h_M \le C_p \eta_M / M^2$, an expected truncated quadratic error of order $d^{-1/2}e^{-I}$ suffices. A bound Ce^{-Hd} is a convenient sufficient condition whenever $I \le C_I d$ and $H > C_I$. The expected hit-mass condition on h_M is weaker and is itself enough.

Proof. For analysis only, replace e_j by $e'_j = e_j \mathbf{1}\{|e_j| \le M\}$. The truncated coefficients satisfy

$$\left| b + \sum_j r_j e'_j \right| \le B + \eta_M / M \le B + V/M, \qquad \sum_j r_j (e'_j)^2 \le V.$$

Hence lemma 2.1 bounds their score error by $C_M d^{-1/2} e^{-I}$ on $\mathcal{E}$. The original and truncated random scores are identical whenever $X_j = 0$ for every j with $|e_j| > M$. Conditional on training, the probability that this fails is at most

$$\sum_{j:|e_j|>M} p_j = h_M \le \frac{C_p}{M^2} \eta_M.$$

Integrate this bound on $\mathcal{E}$ and pay at most $\mathbb{P}(\mathcal{E}^c)$ outside it. No coefficient truncation is an additional input or computation required of the implemented decoder. □

For the degree-floor projection there is a concrete reduction to a weighted row count. Define

$$\zeta_j = \sqrt{\frac{nd}{G_j}} \, \|U_{j\cdot}\|_2 .$$

If $e = (H - I)f$, $\|f\|_\infty \le C_f$ and $\sum_j G_j \le C_G nd$, then deterministically

$$|e_j| \le \frac{\|U_{j\cdot}\|_2}{\sqrt{G_j}} \left\| U^\top G^{1/2} f \right\|_2 + C_f \le C_f \sqrt{C_G} \, \zeta_j + C_f.$$

Choosing $M = C_f \sqrt{C_G} L + C_f$ thus gives $\{|e_j| > M\} \subseteq \{\zeta_j > L\}$. Since $p_j \le Cd/n$, the sufficient expected-count estimate

$$\mathbb{E}\left[\mathbf{1}\{\mathcal{E}\} \sum_j \mathbf{1}\{\zeta_j > L\} \right] \le Cne^{-Hd}, \qquad H > C_I, \tag{154}$$

would imply $\mathbb{E}[\mathbf{1}\{\mathcal{E}\} h_M] \le Cde^{-Hd} = o(d^{-1/2}e^{-I})$ whenever $I \le C_I d$. Equation (154) is the required new spectral tail estimate, not a conclusion of proposition 9.1.

9.6 What is still required for the actual sparse likelihood theorem

The preceding propositions prove more than spectral consistency: the weighted projection has bounded tilted mean and variance, and the sharp tail permits either integrated exponential coefficient control or an expected truncated quadratic bound. They do not yet prove either of those final coefficient-tail hypotheses for the actual floor eigenvectors.

Three coupled estimates remain. First, one must establish the needed coefficient tails on the recomputed deleted graphs. A sufficient uniform condition would be $\|U_{j\cdot}\|_2 / \sqrt{G_j} \le C/\sqrt{nd}$, but lemma 9.3 permits rare violations. Second, the actual full-graph score must be compared to this independent deleted-graph score, including the changing floor and the degree correction in (151). The dense-range LOO lemma used a uniform row bound and cannot be invoked unchanged. Third, initialization and estimated-parameter fitting must be controlled with exceptional vertices included. Aggregate spectral error alone does not supply the required fitted likelihood remainder.

For $d = o(\log n)$ the already proved n^{-r} geometry failure is negligible relative to $d^{-1/2}e^{-C_I d}$ for any fixed $r > 0$. The unresolved issue is therefore not the probability of the regularized norm event. Nor should one declare failure of the entire graph whenever any vertex has an unusually small degree. The truncated-score comparison shows how local exceptional coordinates can instead be paid when the tested row encounters them.

Zhang's spectral clustering proof [5, Theorem 3 and Corollary 1, p. 7326] establishes exponentially small expected truncated row energy and exceptional-row counts for the trimmed adjacency eigenspace. That result provides a concrete model for (154); its eigenspace and normalization differ from the floor-normalized DC construction here. Its final leading-exponent conclusion does not, by itself, imply the bounded perturbation cost and sharp prefactor required here. The intended nontrivial sparse target on the present fixed-separation, bounded-theta class is $d = np \to \infty$, together with the stated Bernoulli remainder condition. A vanishing signal gap or degree parameters approaching zero requires additional quantitative work. We have not replaced these outstanding estimates by an assumption that the desired constant risk ratio already holds.

References

[1] Emmanuel Abbe, Jianqing Fan, Kaizheng Wang, and Yiqiao Zhong. Entrywise eigenvector analysis of random matrices with low expected rank, 2017. arXiv:1709.09565; Lemma 7, equation (C.2), version accessed 20 October 2026.

[2] Afonso S. Bandeira and Ramon van Handel. Sharp nonasymptotic bounds on the norm of random matrices with independent entries. *The Annals of Probability*, 44(4):2479–2506, 2016. Corollary 3.12 and Remark 3.13; arXiv:1408.6185v4.

[3] Chao Gao, Zongming Ma, Anderson Y. Zhang, and Harrison H. Zhou. Community detection in degree-corrected block models. *The Annals of Statistics*, 46(5):2153–2185, 2018. Local testing formulation checked in the author manuscript arXiv:1607.06993.

[4] Can M. Le, Elizaveta Levina, and Roman Vershynin. Concentration and regularization of random graphs. *Random Structures & Algorithms*, 51(3):538–561, 2017. Theorem 2.1 and Remark 2.2; arXiv:1506.00669.

[5] Anderson Ye Zhang. Fundamental limits of spectral clustering in stochastic block models. *IEEE Transactions on Information Theory*, 70(10):7320–7348, 2024.

[6] Zhixin Zhou and Ping Li. Rate optimal Chernoff bound and application to community detection in the stochastic block models. *Electronic Journal of Statistics*, 14(1):1302–1347, 2020.